OCVR, the evidence behind each rule: a literature check of the variable-rate economics method

Olds College — OFPE / OCVR variable-rate economics

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Olds College OFPE team. Literature review supporting the OCVR app (Olds College Variable-Rate economics, version 1.1.0), which evaluates on-farm seed-rate and nitrogen-rate trials by NDVI zone and answers whether variable rate would have paid against a flat rate. It was checked on three on-farm trials of 2026.

Prepared 29 October 2026. Citation keys are first author and year, as in Section 12 and `references.csv`*; same-author, same-year works carry a letter (Harker 2017 = canola seeding rate and seed size, Harker 2017a = nitrogen and seeding rate versus novel inputs; Mangin 2022 = part I yield, Mangin 2022a = part II grain nitrogen). Every statement about a paper comes from the verified evidence extracted from it (*`evidence_by_rule.json`*), and every reference says how far it could be read (full text, abstract or publisher preview, not read). Appendix A gives the method of the check.*

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1. Executive summary

What was checked. The method of the app was written out as 28 rules (zones Z1–Z4, join J1–J4, response R1–R5, economics E1–E11, treatment strips T1, prices and Prairie context P1–P2, the cleaned yield map C1), each with its parameters as coded. For each rule, every cited paper was read (the full text where it was open, otherwise the abstract or the publisher's preview) and judged on whether it says what the rule needs. Each judgement was then contested by a second, independent reader with the paper in hand, and a final critic listed the topics with no reference at all. 112 rule–paper pairs were judged: 16 support the rule as implemented, 69 support part of it, 20 do not support it, and 7 could not be read far enough to judge. Of the 28 rules, one is well supported (E11, one season is ex post), 21 are partly supported, and 6 are house conventions with no published counterpart (J4, R4, R5, E8, E10, P1). 69 further references were found and confirmed, of which 63 were adopted (the other 6 were already in the earlier base), plus 14 for topics nothing cited (the Sentinel-2 product and its cloud mask, the NDVI itself, the Prairie NDVI–yield window, within-field variability, seeding-rate responses, the cost of not cleaning). All 147 DOIs of the final list resolve on Crossref or OpenAlex with a matching title, first author and year.

What the literature supports. The frame of the app is the published one: the partial budget and the value of information of Swinton & Lowenberg-DeBoer (Swinton 1998; Bullock 2002); the temporal-mean zoning of Blackmore 2000 carried to satellite NDVI by Lenoir 2024 (Sentinel-2) and Basso 2019 (Landsat); on-farm precision experimentation with randomised plots, buffers at the plot ends and footprint overlay (Bullock 2019; Trevisan 2021; dos Santos 2025); a parametric response with the plateau form reported beside the quadratic (Cerrato 1990; Bullock 1994; Miguez 2022); standard errors clustered by the randomisation unit (Cameron 2015; Abadie 2023); the in-sample value of VR being biased upward by construction and the remedy of judging it on data that did not choose it (Mieno 2024; Chernozhukov 2018; Athey 2016); a minimum worthwhile gain because payoff functions are flat near the optimum (Pannell 2006; Pannell 2019); and the ex-post nature of one season's optimum (Rajsic 2008; Bullock 2020; Puntel 2016).

What the literature does not support, and what we did about it. The previous literature base (27 October) attributed to Mieno 2024 a zone × rate test and a cross-fitting procedure that the paper does not contain, to Bullock 2019 a grower-compensation computation it does not describe, to Bullock 2000 a sentence that is not in its abstract, and to Lee 2012, Gauci 2024, Malhi 2001 and Mangin 2022 findings they do not report; its table of defaults named a 5-fold cross-fit, 500 resamples, 300 observation units per zone, a 50 % footprint share and a 4-SD filter that the app did not have (its bootstrap default was then 100). Those attributions are corrected here and in the README and the PDF report. The check also changed the app in four places (Section 11): Wald tests are now judged against F(h, plots − 1) rather than chi-square, because the chi-square reference over-rejects with few clusters (Cameron 2015); the bootstrap default went

from 100 to 500 resamples (1,000 advised for a report to publish, a house choice; Hernandez 2008 found that bootstrap intervals on the optimum "showed great potential" and Jaynes 2011 used 1,000 Monte Carlo realizations); the label "DIFM grower compensation" became "the ex-post cost of the trial to the grower" (Puntel 2024); and the advice to cut a rate now carries the warning that under-application costs more than over-application when the true response is steeper than estimated (Babcock 1992; Rajsic 2008).

House choices. The following have no published counterpart and are presented as calibrations, not findings: the median of the three highest-NDVI dates of a 15 June – 15 August window, the 3 × 3 smoothing, the equal-area tertiles and the 0.5 ha patch rule (Z1, Z3); the 95 % footprint share and the unit thresholds of 10 records and 150 m² (J2, J3); reading a trial design back from the as-applied map, with its separation score, shape tests and mixed-plot rule (J4); the merge rule of 20 units and 3 design rates per zone (R4); the zone × rate Wald test as one of the two conditions for VR, beside the cross-fitted gain (R5); the thresholds $3/ac, 70 % of repeats and $p < 0.10$ and their AND logic (E4, E5), calibrated on one synthetic control; the one-sided linear extension of a status quo outside the trial range and its 25 % cut-off (E9); the ± 30 % price grid (E7); the corner-share rule for the NDVI prescription (E10); and the realised-trial accounting with every input at its status quo (E8). Each is a defensible choice validated on the app's synthetic trials and the three worked cases, without outside validation.

Open points. Seven rule–paper pairs could not be judged: Grant 2012 could not be read at all; Malhi 2001 could not be read for P2 (its abstract and introduction, read for P1, show nothing on the N plateau); and Taylor 2007, Bullock 2000 (R2), Bullock 2019 (J4), Piepho 2011 and Lee 2012 were read only at abstracts that do not settle the rule. Blackmore 2000 is closed: its abstract and introduction snippets were read on the ScienceDirect preview, and its equations only as Lenoir 2024 reproduces them. Three published results run against the zone method and are stated as limits rather than resolved: Georgi 2018 discards exactly the saturated high-NDVI images the window prefers; Lenoir 2024 found the NDVI–yield correlation of Belgian winter wheat falling after the end of May and chose its image with the yield map in hand; Blackmore 2003 found multi-year trends washing out on four English fields. No published NDVI study recovers unstable zones with a temporal CV (Lenoir 2024: sensitivity zero). For the Prairies the only timing evidence located is at MODIS scale (Mkhabela 2011), and it supports the window.

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2. How to read this document

Sections 3 to 9 go through the 28 rules in the order of the app's steps. Each rule has four parts: *what the app does* (with the parameters as coded on 29 October 2026), *what the literature says* (the specific finding, method or number in each paper that bears on the rule, with the key of the paper and how far it was read), *support* (well supported, partly supported, or house choice, with the house choices named), and *what changed after the check*. Section 10 lists the house contributions in one place, Section 11 the changes made to the app and its documents, Section 12 the verified reference list. Appendix A describes the check itself.

The support levels mean: **well supported**, every element of the rule, including its numbers, is found in a paper that was read; **partly supported**, the approach is published but one or more of the rule's parameters, thresholds or specifics are the app's own; **house choice**, no paper read describes the rule, and the citations in the earlier literature base supported at most the premise.

3. Zones from multi-year NDVI

OCVR splits the field with a multi-year Sentinel-2 NDVI map, not with the trial-year yield map, because the zones must be a prior that exists before the trial and does not depend on its outcome. Four rules build that map: how one season is summarised (Z1), which seasons count (Z2), how the temporal mean becomes zones (Z3) and what the zones are allowed to mean (Z4). The published precedents are yield-map trend analyses (Blackmore 2000; Blackmore 2003) and their satellite ports (Georgi 2018; Lenoir 2024; Basso 2019). They support the skeleton of the method: one layer per season, normalised to the field mean, averaged over at least three seasons, with a dispersion statistic. They do not support the specific settings the app ships with, and three of them report findings that cut against the premise. Those are stated below, rule by rule, with how each paper was read (full text, abstract, or not read), as the cite-check of 2026-10-29 requires.

Z1. One NDVI layer per season and the multi-year temporal mean

What the app does. For each past season (the trial year left out) the app takes, inside a window of 15 Jun to 15 Aug, the three clearest Sentinel-2 L2A dates with the highest field-mean NDVI, computes their per-pixel median, divides it by that season's field mean and averages the seasons into a temporal mean, with a temporal CV alongside. Clear dates come from the L2A scene classification.

What the literature says. The method is Blackmore's: six years of yield maps on a 7-ha wheat and oilseed-rape field classed as high-stable, low-stable and unstable (55, 45 and 0 % of the field for the single crop; 58, 39 and 3 % with both crops). The equations could be read only as reproduced in Lenoir 2024, the full text being closed (Blackmore 2000, abstract and snippets), and nothing in it concerns dates. Lenoir 2024 (full text) ports it to Sentinel-2 L2A on 35 Belgian wheat fields (2016-2021, 1 Apr to 30 Jun, 5-7 images a year): three years minimum, median better than mean (71 against 69 % correct), but the NDVI-yield correlation fell after the end of May, the second half of May was chosen rather than peak canopy, the one map per year was picked with the yield map in hand, and the CV never recovered an unstable zone (sensitivity 0). Georgi 2018 (full text), on RapidEye NIR, discards dense-canopy images (mean NDVI above 0.73 in the text, 0.78 in the figure), the saturated images Z1 prefers, and averages every suitable image of a year. Blackmore 2003 (abstract) found that the spatial variability of four fields "cancelled out over time" and advised managing on the current season. Ali 2019 (full text) found wheat correlations above 0.70 only before heading; Boydell 2002 (abstract) needed five or more years. Basso 2019 (methods, full text) is the closest satellite analogue, one late-July Landsat composite per season relative to the field mean, but its 0.15 temporal SD is computed on the binary high/low class, a never-switches rule, not a CV. Košánová 2025 (abstract; method from a poster) and Holben 1986 (abstract) support, respectively, one peak composite per season and the highest-NDVI date as a cloud filter.

Support. Partly supported: the per-season layer, the normalisation and the multi-year mean have precedents; the median of the three highest-NDVI dates, the 15 Jun to 15 Aug window and a CV without Blackmore's 30 % threshold have none, and no NDVI study recovered unstable zones with a CV.

What changed after the check. The three contradictions are now stated rather than cited as support. Product references were added: Drusch 2012 (mission), Main-Knorn 2017 (Sen2Cor and the SCL mask), Skakun 2022 (cloud masks compared; why the median of three dates stays as a residual-cloud safeguard), Tucker 1979 (NDVI) and Hunt 2019 (wheat yield at 10 m on 39 UK fields, RMSE 0.66 t/ha). The window now cites Mkhabela 2011: MODIS NDVI 2000-2006 predicted Prairie barley, canola, pea and spring wheat yield best from the third dekad of June to the third dekad of July (sub-humid) or the first dekad of July to the first of August (semi-arid and arid), at crop-district scale, not field scale; Chipanshi 2015 is secondary support. All were confirmed from abstracts and metadata, not full text.

Z2. Which seasons enter the history

What the app does. Seasons of the trial's crop are used when at least three are known, then the crop group (wheat, barley, durum, oats; only when it holds three seasons and the trial crop belongs to it), then every clear season, and the report says which. Three is the threshold for that fallback, not a requirement: a run with fewer seasons in total is neither stopped nor separately flagged. The trial year is always excluded.

What the literature says. Three seasons is Lenoir's algorithmic minimum and it bound often: six of 35 fields had them (Lenoir 2024, full text). Georgi 2018 (full text) found three years enough where crop patterns are strong and stable and asked for five or more on a dynamic field (its 3-versus-6-year test ran on the calmer fields); Boydell 2002 (abstract) also needed five or more. Diker 2004 (abstract) is a precedent for three: a two-state frequency analysis of three corn seasons on two Colorado fields, 17 and 6 % of the fields consistently below the mean, 43 and 42 % above. Maestrini 2021 (full text) found modest gains from more years: going from six to eleven or twelve (the paper gives both) cut uncertain pixels from 55 to 50 %. The same-crop preference comes from Blackmore 2000 (abstract), where single-crop maps were more stable than mixed-crop ones, and from Joernsgaard 2003 (abstract): on 82 fields the between-year R^2 of the yield pattern ran from 0 to 0.81 (mean 0.27), highest for similar crops, and trend maps are expected to explain no more than a third of annual within-field variation. Georgi 2018 supports it only indirectly (the segmentation "does not validate well with canola yield maps") and Maestrini 2021 only in a hedged sentence about wheat. Schepers 2004 (abstract) is a caution: soil-based zones on irrigated corn separated yield by about 25 % in three of five seasons and by 5 % or less in the driest and wettest. Blackmore 2003 (abstract) warns that a multi-year trend can wash out.

Support. Partly supported: the three-season floor, the same-crop preference and the trial-year exclusion have precedents; the cereal-group fallback between "same crop" and "all seasons" has none (Blackmore 2000 compares one crop against all; Joernsgaard 2003 speaks of "similar crops"), and no paper tests three as an accuracy threshold.

What changed after the check. The trial-year exclusion is now an application of published practice, not a house choice: Lenoir 2024 kept only seasons with uniform N fertilisation and Georgi 2018 dropped 2012 because the field was split between wheat and sugar beet. The defaults table attributed a z-score normalisation to Georgi 2018, Lenoir 2024 and Ali 2019; for the delineation step that is wrong (Lenoir standardises only for the Moran's I check; the delineation divides by the annual field mean) and section C is rewritten. Joernsgaard 2003, Diker 2004 and Maestrini 2021 are added to this rule. This review states three as a floor and, after Georgi 2018, prefers five or more where the pattern changes between years; the app's season caption says only "at least 3", and neither the app nor its PDF report mentions five or more yet.

Z3. From temporal mean to zones

What the app does. The temporal mean is smoothed with a 3 x 3 window on the 10 m grid, cut into k equal-area quantiles (tertiles by default; k-means as an option) and patches under 0.5 ha are merged into their neighbour. k is 3 by default and may be 2 to 5.

What the literature says. This is the weakest-supported rule of the group. Its anchor, Blackmore 2000, could not be read in full on this check (Elsevier full text closed, no open copy; the Z3 checker found no deposited abstract, but the Z1 and Z2 checkers read the abstract on ScienceDirect and in Consensus); what is known of its method comes from that abstract and from Lenoir 2024, and its classes come from mean and CV thresholds, not quantiles, with no smoothing window, patch size or k-means. Taylor 2007 is unread beyond an abstract that gives no number of classes, algorithm or minimum zone size, so the claims hung on it (2-4 classes; a minimum zone tied to implement width) stay unverified. Gavioli 2019 (abstract)

compared 20 clustering algorithms on six years of soybean and maize yield and found McQuitty's method and Fanny best; the earlier summary that it favoured k-means is wrong. Fridgen 2004 (abstract) is MZA: fuzzy c-means with FPI and NCE indices that suggested two or four zones on claypan fields; it supports choosing k by an index, not tertiles. Milne 2012 (abstract) supports smoothing for compact zones on three years of wheat yield, and found N-response curves parallel across yield-based zones. The analogues of the smoothing step are Ping 2003 (abstract), where a post-classification filter of about 30-60 m removed fragmentation while keeping class means and grids coarser than 16 m lost the pattern, and Javadi 2022 (full text), where islands below a user-set size are relabelled by majority rule because machine actuators cannot follow them, the size matched to the control unit, with k = 4 run without a rationale. Dobermann 2003 (abstract) is partly contrary: empirical three- or four-class splits of irrigated maize yield explained under 54 % of the variance, six or seven cluster classes on the mean 60-66 %, and clustering the temporal mean beat mean-plus-SD or all years.

Support. House choice: equal-area tertiles, the 3 x 3 window before classification (Ping 2003 filters after; 3 x 3 on 10 m pixels is the low end of their 30-60 m), the 0.5 ha patch (no paper gives a size) and k = 3 with 2-5 allowed (Fridgen 2004 two or four; Javadi 2022 four; Dobermann 2003 six or seven for irrigated maize) have no published counterpart; clustering the temporal mean and smoothing for compactness do.

What changed after the check. The defaults table said the minimum zone was 5 % of the field or three combine widths while the code uses 0.5 ha; section C is rewritten to the code's defaults and 0.5 ha is labelled a house value. Tertiles are presented as a practical choice, not a variance-optimal one, with Dobermann 2003 as the contrary evidence. Leroux 2019 (abstract) is added for whether a field is variable enough for zones, and Maestrini 2018 (abstract) as the published reason an unstable class exists and why it follows weather and topographic wetness. Blackmore 2000 is marked as not read for Z3 (second-hand; its abstract was read for Z1 and Z2) and Taylor 2007 as read only at abstract level; obtaining both through the college library remains open.

Z4. What the zones are allowed to mean

What the app does. Zones are expected to rank yield level, and the R^2 of unit yield by zone is reported; they are not expected to carry the response to the rate, because peak NDVI saturates in dense cereal canopies. Whether the response differs by zone is left to the zone x rate test and the cross-fit, and NDVI is never turned into an N recommendation directly.

What the literature says. The separation of yield potential from response is the best-supported part of the chapter. Franzen 2016 (full text, open access) states it as a core idea, "Yield potential and N response are independent": the OSU algorithm needs separate estimates of each, sensors "are unable to quantify the amount of excess N in plants", and plants showing no N stress mid-season often still respond, so Missouri built in a recommendation floor. Its source is Lory 2003 (abstract): on 298 corn N-rate experiments, yield at the optimum explained 2 % of the variation in the economic optimum ($r^2 = 0.02$), delta yield 47 %. Raun 2002 and Raun 2005 (abstracts) are the proximal origin, an in-season yield estimate from NDVI and the response from an N-rich strip, NUE up by more than 15 %; nothing in them concerns satellites, multi-year zones or saturation. Colaço 2018 (abstract and introduction): sensor-based N management saved 5-45 % N with little yield effect, averaged US$30/ha profit with losses in about 25 % of studies, and no algorithm was reliable across conditions; it supports only the "never a direct N recommendation" clause. Schepers 2004 (abstract) supports "zones rank level, unstably": 25 % between zones in three of five seasons, 5 % or less in the others. Saturation is covered by Erdle 2011 (abstract, highlights, introduction): NDVI and related indices "tended to saturate with increasing crop stand density" in wheat above a LAI of 2-3, on proximal sensors, showing as lost discrimination between N levels rather than a low r^2; and by Mutanga 2004

(abstract): standard NDVI R^2 = 0.26 on dense grass against 0.77-0.80 for red-edge indices. Scharf 2006, as summarised by the completeness check, found yield level explaining 15 % of the optimum N rate, with VR by optimum worth $38/ha against $2/ha by yield.

Support. Partly supported: that yield level and response are different quantities is well documented; the transfer of that in-season proximal logic to multi-year satellite zones is the app's own analogy, and the statement that peak Sentinel-2 NDVI saturates in Alberta cereals, with a 3-5 % CV of normalised NDVI, is an OCVR observation with Erdle 2011 for the mechanism only.

What changed after the check. The zone x rate test and the cross-fit are no longer attributed to Mieno 2024, which is about geographically weighted regression overstating the value of VR and is now cited for the in-sample bias only. The test Z4 hands over to is judged against F(h, plots - 1) rather than chi-square, after Cameron 2015, because with few plots per zone the cluster-robust Wald test at 0.10 over-rejects. Lory 2003, Erdle 2011 and Mutanga 2004 are added; Hunt 2019 and Chipanshi 2015 (both abstract) are the field-scale and Prairie evidence that NDVI tracks cereal yield at all.

4. The cleaned yield map

C1. The yield map enters already cleaned

What the app does. The yield map is cleaned upstream by OCCD (a cascade of filters: delay, start and end of pass, moisture, speed, overlap) and OCVR removes no further record on its yield value. Records leave OCVR only through the containment rule of the join (J2), the unit-size rule (J3) and the no-zone rule, all counted in the attrition table.

What the literature says. Sudduth 2007 (full text) is Yield Editor: twelve filters (DELAY, START, END, MAXV, MINV, SMV, MINS, MAXY, MINY, STDY, POS, MAN), the delay set per field by matching transitions of transects harvested in opposite directions (9 s leaves an offset, 14 s makes the pattern contiguous), 12.6-26.9 % removed on five fields, and a suggested sequence in which DELAY, START, END and MAN are needed on any dataset. The words "moisture" and "overlap" do not occur in it; partial swaths are left to the operator-dependent MINS filter (0.5 % removed) and manual deletion. Lyle 2014 (abstract; full text closed) names the four error classes (harvester dynamics, continuous measurement of moisture and yield, position, operator) and calls for extensions to speed, narrow finishes, turns and overlaps; it is the only original citation that mentions moisture and overlap. Arslan 2002 (abstract) supports only the premise that errors exist and some can be filtered; the 10-40 s filling time taken from it elsewhere is not in the abstract and is unverified. Simbahan 2004 (abstract) is the citation for a moisture screen and pass-end trimming, with a correction: the mandatory primary screening is header status and pass-end delays (72-85 % of the 13-20 % removed); moisture, flow and distance outliers are step 3 of the secondary screening. Vega 2019 (abstract) removes about 30 % of 595 datasets in a fixed order, a third as spatial outliers. Leroux 2018 (abstract) detects filling and emptying, speed changes and partial cutting bar before further processing. Kharel 2019 (abstract): on 145 silage and 88 grain fields, uncleaned, automatically cleaned and Yield Editor maps give different yields and manual cleaning differs from person to person, the argument for one fixed protocol upstream.

Support. Partly supported: cleaning as a separate, fixed-order pre-processing stage is well documented; the second half of the rule, that a cleaned map is not filtered again downstream, has no published counterpart and is a house convention.

What changed after the check. The 10 m trim of the join no longer claims to cover the yield-monitor delay: the delay is corrected upstream by OCCD, so the trim covers the applicator's rate change and the residual

delay error, with DIFM's ~10 m buffers (Bullock 2019 p. 2743) and Trevisan 2021 as precedents; delays now cite Chung 2002, which measured 10-23 s varying within a field. The defaults table listed a 4-SD outlier filter on unit means that does not exist in the code; section C is rewritten without it. The check found the rule incomplete as a description of the code, since units under 10 records or 150 m² and pieces that receive no zone are dropped too; the rule text above now says so. Citations are split by content: Sudduth 2007 for delay, pass ends and speed, Simbahan 2004 for moisture and pass ends, Lyle 2014 and Leroux 2018 for overlap and partial swath, Kharel 2019 (added) for why cleaning happens once, upstream.

Rule	Support	Verified references (keys)	House choices
Z1	partly supported	Blackmore 2000 (abstract and snippets; equations via Lenoir 2024), Blackmore 2003 (abstract; contradicts), Georgi 2018 (full text; contradicts the high-NDVI preference), Lenoir 2024 (full text; contradicts the window and the date choice), Ali 2019 (full text), Boydell 2002 (abstract), Basso 2019 (methods full text), Košánová 2025 (abstract and poster), Holben 1986 (abstract); added: Drusch 2012, Main-Knorn 2017, Skakun 2022, Tucker 1979, Hunt 2019, Mkhabela 2011, Chipanshi 2015 (abstract or metadata)	median of the three clearest highest-NDVI dates; window 15 Jun to 15 Aug; temporal CV without the 30 % threshold
Z2	partly supported	Lenoir 2024 (full text), Georgi 2018 (full text), Blackmore 2000 (abstract), Blackmore 2003 (abstract), Boydell 2002 (abstract), Schepers 2004 (abstract); added: Joernsgaard 2003 (abstract), Diker 2004 (abstract), Maestrini 2021 (full text)	the cereal-group fallback; three seasons as the same-crop threshold (Lenoir's algorithmic minimum)
Z3	house choice	Blackmore 2000 (not read; second-hand), Taylor 2007 (unread beyond abstract), Gavioli 2019 (abstract), Fridgen 2004 (abstract), Milne 2012 (abstract); added: Ping 2003 (abstract), Javadi 2022 (full text), Dobermann 2003 (abstract; partly contrary), Leroux 2019 (abstract), Maestrini 2018 (abstract)	equal-area tertiles; 3 x 3 smoothing before classification; 0.5 ha minimum patch; k = 3 default, 2-5 allowed
Z4	partly supported	Franzen 2016 (full text), Lory 2003 (abstract), Raun 2002 (abstract), Raun 2005 (abstract), Colaço 2018 (abstract and introduction), Schepers 2004 (abstract), Erdle 2011 (abstract and introduction), Mutanga 2004 (abstract), Scharf 2006 (abstract, via the completeness check); Mieno 2024 for the in-sample bias only; Cameron 2015 for the F reference	transfer of the proximal yield-potential-versus-response logic to multi-year satellite zones; "peak Sentinel-2 NDVI saturates in Alberta cereals, CV 3-5 %" as an OCVR observation
C1	partly supported	Sudduth 2007 (full text), Lyle 2014 (abstract), Arslan 2002 (abstract), Simbahan 2004 (abstract), Vega 2019 (abstract), Leroux 2018 (abstract), Kharel 2019 (abstract), Chung 2002 (abstract), Bullock 2019, Trevisan 2021	no downstream filter on the yield value after OCCD; records leave only through the containment, unit-size and no-zone rules

5. Assigning yield records to plots and zones

Rules J1 to J4 take the cleaned yield map (OCCD) and the trial design and turn them into the analysis units the response models use. Every reference states how it was read (full text, abstract only, or not read); bracketed numbers are those of section B of `LITERATURE.md`.

J1. Each yield record is an area, assigned to a plot by overlay

What the app does. Every cleaned yield-monitor record becomes a rectangle, swath width × distance travelled, centred on the logged point along the heading corrected to grid north (`yield_polygons`, `assign_yield`). The rectangle is overlaid on the trimmed plots and the record is assigned by the area inside a plot, not by the position of the point. A point mode (point-in-polygon) is used when the file lacks a heading, a distance or a swath; only when the swath or the distance is missing does it flag that unit areas are counts, not m².

What the literature says. The direct counterpart is dos Santos 2025 (full text). Its "simple" method averages the observations that fall within an experimental unit (point-in-polygon); its RITAS method builds polygons from the swath and the distance between consecutive measurements, resolves overlaps by keeping the first measurement, and tracks passes that would collect grain from two units. In 2,000 simulated trials (120 units, 6 N rates) the RMSE of the agronomic optimum was 1.5 kg N/ha with polygons against 5.1 with points, that of the economic optimum 1.0 against 3.1, and points biased the economic optimum by −7.9 kg N/ha against 0.1; on a strip trial the optima were 157.0 and 133.5 kg N/ha, on a checkerboard seed trial 118.7 and 92.8 × 10³ seeds/ha. The authors blame under-estimated yield where passes overlap and dilution between adjacent units. Caveats: the comparison bundles cleaning, assignment and block kriging, and units one swath wide cannot shed crossing polygons. Drummond 1999 (abstract only) derives each record's harvested area from the GPS trajectory, improving harvest-area estimates by 11 % over a constant full swath; Han 1997 (abstract only) finds effective cut widths of 1.18–2.24 m against a nominal 1.52 m but collapses units back to centroids, a precedent for the area representation, not for overlay. Bullock 2019 (full text) does not support the method: DIFM treats readings as points, 160 per 24.4 × 88.5 m plot, gathered into subplots of about 20 m; it only shares the recognition that a swath can straddle two plots, handled by removing the whole westernmost pass of each plot. Trevisan 2021 (full text) averages the points inside 37–87 m² polygons. Sudduth 2007 (abstract only; closed access) is a cleaning tool that deletes single records and says nothing about assignment.

Support. Partly: the area representation and the cost of point assignment are published (dos Santos 2025, Drummond 1999, Han 1997); the rectangle centred on the logged point, the direct overlay on plots rather than a fine grid, and the reliance on an upstream-cleaned map are house choices.

What changed after the check. The overlay now cites dos Santos 2025 instead of Bullock 2019 and Sudduth 2007; Bullock 2019 is kept for J2 and stated as point-based; Sudduth 2007 moves to C1; Drummond 1999 and Han 1997 are added. The section C table of `LITERATURE.md` said "drop footprints < 50 % inside a plot" against the implemented 95 % and is corrected.

J2. Trimmed plot ends and the 95 % containment rule

What the app does. Each plot is shortened by 10 m at both ends of its long axis (`trim_plots`; side trim optional, default 0 m). The trim is taken off the plot polygon itself, so a wedge or trapezoid clipped to the boundary loses its full 10 m along the whole end edge; a plot shorter than 21 m keeps a 1 m strip so its records are counted and then dropped by the unit rule. A record is kept only when at least 95 % of its footprint lies inside one trimmed plot; straddling records are dropped and counted (`min_share = 0` switches to clip-and-weight).

What the literature says. Bullock 2019 (full text, p. 2743) is the precedent for the trim: DIFM places "approximately 10 m-long transitional buffer zones" at the ends of each plot because yield monitors read with a delay between yield zones and applicators lag at treatment boundaries; DIFM does not use buffer data; plots are 75–95 m long. Kyveryga 2018 (full text, p. 199) requires a clean pass within one treatment and removes the headlands and about the first 50 ft of each pass, a start-of-pass trim only. Alesso 2021 (full text, p. 1370) sets a minimum plot length of 36 m, about 18 s, for actuator, threshing and monitor delays, in a simulation that assigns no records. Chung 2002 (full text, Tables 3–4) measured yield delays of 12–16, 15–23 and 10–17 s in three fields, moisture delays of 14–23 s, and differences of 1–3 s between regions of one field; a wrong delay blurs region edges into a saw-tooth, so transition records stay unreliable after one correction. Lark 2003 (abstract only) accounts for the combine's smoothing of the mass-flow signal. Trevisan 2021 (full text, p. 350) is the alternative: plots long enough for at least 30 s of harvest, transition points kept. Gauci 2024 (full text, p. 774) explicitly does not investigate the minimum length over which a

monitor detects a yield difference and contains no lag, delay or transition content; it reports under 3 % error on strip totals inside the calibration range and over 15 % outside. Lee 2012 (abstract only; no open copy) proposes phase-correlation delay identification, accurate in about 90 % of test sets, with no delay magnitude. No paper contains the 95 % share.

Support. Partly: the 10 m trim is DIFM's published buffer length and the one-treatment rule has precedents, but the 95 % footprint share is a house choice (dos Santos 2025 apportions a crossing polygon proportionally; Bullock 2019 uses subplots inside the buffered interior).

What changed after the check. Delay figures now cite Chung 2002, 10–23 s and varying within a field, instead of Lee 2012, kept only as a delay-identification method; Gauci 2024 is dropped. The old arithmetic "10–15 s × 1.5–2.5 m/s = 15–35 m" applied to an uncorrected delay: OCCD corrects the delay upstream, so the trim covers the applicator's rate change and the residual delay error, with Bullock 2019 and Trevisan 2021 as precedents. The code docstring's attribution of the 95 % share to DIFM is not in Bullock 2019; it is a house choice.

J3. Analysis units: plot × zone, with floors on records and area

What the app does. The kept record pieces are cut by the NDVI zone and grouped into units of one plot inside one zone (`build_units`; a `plot` mode without zones exists). A unit needs at least 10 records and 150 m² (the area test is skipped when the file gives no real area per record); unit yield is the area-weighted mean of its pieces; the applied rate is the median of the as-applied points inside it (at least 5 points, otherwise the plot median), or the design rate when there is no map; a piece on no zone takes the zone of the nearest valid cell among its 8 neighbours, and is dropped and counted when there is none. The plot, the unit of randomisation, is the cluster for every standard error downstream.

What the literature says. Bullock 2019 (full text) takes data from the middle 50 % of each plot and partitions the interior into three or four subplots of about 20 m (19 m in its Fig. 7), the observational unit for yield, moisture, as-applied and as-planted rates and soil type; it then tests whether SSURGO zones respond differently and classifies areas by GWR. It describes the aggregation only as a "resampling process" and never names the statistic; per-subplot counts sometimes quoted for it are arithmetic on its example, not printed values. Trevisan 2021 (full text) subdivides each plot into 10–20 polygons of 37–87 m², uses the average of the points in each (2,816–3,276 units per field) and the recorded rate after a 3 × IQR filter. Cho 2021 (full text) runs a single strip (9 m in grain fields, 24 m in dairy fields) across at least two yield-stability zones and puts the zone into the model (`Yield ~ Zone + Nitrogen:Zone + …`) on 553–5,220 raw points per field; in one field the N response was +0.25 Mg/ha in one zone and −0.25 in the other. Marchant 2019 (full text) adds zone × treatment interactions to a mixed model on 4,524 cleaned points and gives the only sufficiency figures found, 100 observations for a variogram and 10 rows of the standard treatment, both whole-experiment requirements. Alesso 2021 (full text) is a simulation on 81 m² pixels with 324–1,296 m² plots and supports only that units of tens to hundreds of m² work. Gauci 2024 (full text) states that its result does not characterise what a monitor can discern within a plot.

Support. Partly: sub-plot observational units and zone-specific responses are published practice, but the plot × zone intersection, the 10-record and 150 m² floors, the median applied rate, the area weighting and the design-rate fall-back are house choices (published units are 37–87 m² in Trevisan 2021 and about 20 m × header width in Bullock 2019; Trevisan 2021 uses the mean and Bullock 2019 names no statistic).

What changed after the check. Gauci 2024 is dropped; the claim that DIFM "averages" its subplots is withdrawn, since Bullock 2019 names no statistic; Trevisan 2021, Cho 2021 and Marchant 2019 are added; the unit thresholds and the median are stated as house choices. No contradiction with any paper was

found; the gaps are absences.

J4. Design levels of a second input read from its as-applied map

What the app does. When a second input has no design file, each plot takes the median of the as-applied points inside its trimmed interior (at least 5 points; `rates_from_asapplied`). For k = 2 … 15 the sorted plot medians are split by an exact one-dimensional partition minimising the within-group sum of squares by dynamic programming (`_partition_1d`), and each split is scored by separation: the smallest gap between adjacent groups divided by the largest spacing inside any group (`rate_levels`). The best k must score at least 2 with no rival within 0.85 of it, its gap must reach a noise floor of max(6 × the precision of a plot median, 2 % of the median), and two shape tests catch levels read as one (`_merged_levels`): an internal jump over 6 × a group's other spacings, or a group wider (p10–p90) than 2.5 × the tightest level. Under 24 plots the required score is 4 at 8 plots, falling to 2.5 at 23, and the design is flagged; otherwise the app refuses and asks for a design file. Plots whose p10 and p90 sit nearest two different levels, or whose spread exceeds half the local step, and in either case 3 × the typical plot spread, are mixed and get no design rate (`_mixed_plots`). On the sweeps of `tests/test_levels.py` no wrong number of levels was returned on designs of 20 or more plots per level, refusing about 1 clean map in 150; 5-plot-per-level designs were read in 90–94 %; a gradient on 8–20 plots slipped through in 2–4 % of cases, always with the small-design warning.

What the literature says. No published method reconstructs a trial design from applicator logs. Trevisan 2021 (full text, p. 350) names the practice, grouping as-applied rates into a few levels "historically" for categorical ANOVA, gives no algorithm and keeps the rate continuous; it pre-filters as-applied points at 3 × IQR (p. 345). Hegedus 2023 (full text, p. 680) uses the applicator's as-applied N as a continuous predictor and never mentions a design file. Colaço 2014 (abstract only) classes 30.6 % of logged points as application errors (10 % off target), about 70 % of them at rate changes or during acceleration, the physical basis of the mixed-plot rule; Fulton 2013 (abstract only) finds applicators within 10 % of target less than 45 % of the time, so the prescription cannot stand for the delivered rate. Fisher 1958 (abstract only) is the minimum-within-group-variance grouping the partition implements and Wang 2011 (full text) its dynamic-programming solution with an optimality guarantee; neither covers the separation score. Bullock 2019 could not be read for this check (abstract only; its full text, read for J1–J3, reports nothing on recovering levels), so it is not cited for J4.

Support. House choice: the premise (read the applied rate, not the prescription) and the partition are published; the separation score and its threshold of 2, the ambiguity and shape tests, the small-design rule, the p10/p90 mixed-plot rule and the plot median are unreferenced calibrations validated only on the sweeps above.

What changed after the check. J4 is recorded as a house contribution with no published counterpart; Fisher 1958 and Wang 2011 are cited for the partition only, since the app does use an exact 1-D partition; Fulton 2013 is cited for the premise only, not for plot medians; Colaço 2014, Fulton 2013 and Trevisan 2021 cover as-applied error and filtering, nothing more.

Rule	Support	Verified references (keys)	House choices
J1 footprint overlay	Partly	dos Santos 2025 (full); Drummond 1999 (abstract); Han 1997 (abstract); Bullock 2019 (full, point-based, cited as contrast); Trevisan 2021 (full, point-averaging); Sudduth 2007 (abstract, dropped from J1)	Rectangle centred on the logged point along the grid heading; direct overlay on plots rather than a fine grid; dependence on an upstream-cleaned map

Rule	Support	Verified references (keys)	House choices
J2 10 m trim, 95 % share	Partly	Bullock 2019 (full, p. 2743); Chung 2002 (full); Kyveryga 2018 (full); Alesso 2021 (full); Lark 2003 (abstract); Trevisan 2021 (full, keep-transitions alternative); Lee 2012 (abstract, delay-identification method only); Gauci 2024 (full, dropped)	95 % footprint share in one plot; trim measured from every end edge; 1 m strip for plots too short to trim
J3 plot × zone units	Partly	Bullock 2019 (full); Trevisan 2021 (full); Cho 2021 (full); Marchant 2019 (full); Alesso 2021 (full, simulation); Gauci 2024 (full, dropped)	Plot × zone intersection as the unit; 10 records; 150 m²; median applied rate; area-weighted means; design rate as fall-back; nearest-cell zone fill
J4 levels from the as-applied map	House choice	Fisher 1958 (abstract) and Wang 2011 (full) for the partition; Trevisan 2021 (full); Hegedus 2023 (full); Colaço 2014 (abstract); Fulton 2013 (abstract)	Separation score ≥ 2 and 0.85 ambiguity; jump 6× and width 2.5× shape tests; score 4→2.5 under 24 plots; p10/p90 mixed-plot rule with the 3× spread floor; plot medians; 5-point minimum

6. The response models

Rules R1–R5 of the house method. Every reference carries its reading status at first mention in each rule: (full) the full text was read; (abstract) only the abstract or publisher record could be read; (unread) the body was not available and nothing beyond the abstract is attributed to it. Numbers attributed to a paper come from the verified evidence extracted from it on 2026-10-29. "What the app does" describes `engine/response.py` *and* `engine/treatments.py` *as they stand.*

R1. Response form — quadratic by default, quadratic-plateau as a sensitivity

What the app does. Yield responds to the rate as a quadratic (centred and scaled inside the model). A quadratic-plateau is always fitted alongside by non-linear least squares and its optimum is printed next to the quadratic one; it is bootstrapped only when it is chosen as the main form (the sensitivity fit is a point estimate). The join point is constrained to the supported design range: at or above the second-lowest design rate (below it the rising segment would be fitted to the as-applied scatter around one rate), and allowed above the highest rate when no plateau was seen. The continuous-NDVI model has no plateau version; asked for one, the app fits the quadratic and says so. A line is the form for two-rate designs, because two points identify no curvature.

What the literature says. Both classical comparisons prefer the plateau. Cerrato 1990 (abstract) fitted five models (linear-plateau, quadratic-plateau, quadratic, exponential, square root) to 12 corn site-years with 10 N rates: all fitted about equally by R^2, but the mean economic optima ranged from 128 to 379 kg N/ha at one price ratio, the quadratic "did not validly describe the yield responses", and the quadratic-plateau described them best. Bullock 1994 (abstract) found a systematic bias of the quadratic at every location-sequence of a 10–12-year Illinois data set with 5 N rates: maximum yield 3–6 % larger, optimum 5–60 % larger, profit lower by $0.61–17.12/ha/yr. Ackello-Ogutu 1985 (abstract) rejected polynomial forms against a minimum-and-plateau form. Neither classic supports the quadratic as default. Miguez 2022 (abstract) qualifies the plateau's advantage: always choosing the quadratic-plateau carries a positive bias (the linear-plateau a negative one), and more rates and replications are needed for precision. Dhakal 2021 (abstract) reviews the forms and reports recent preference for stochastic plateaus. "The plateau optimum is less price-sensitive" appears in no abstract read; it follows algebraically (the optimum is capped at the join point) and is stated as reasoning.

Support. House choice for the quadratic default; well supported for always fitting and reporting the plateau and for the warning that the quadratic overestimates the optimum when the response plateaus.

What changed after the check. The docs said the plateau was the default; the code defaults to the quadratic, and the docs now say so. The quadratic stays the default, a house choice taken against Cerrato 1990 and Bullock 1994, because (a) the plateau has its own positive bias (Miguez 2022), (b) the zone and continuous-NDVI interaction models need one parametric form for the rate × zone and rate × NDVI terms, and (c) both optima are printed side by side. Cerrato 1990's model list is corrected (no plain linear model; the fifth is the exponential). Miguez 2022, Dhakal 2021 and Ackello-Ogutu 1985 are added; the price-sensitivity clause is labelled reasoning.

R2. Three structures — one field curve, one curve per zone, a curve that moves with NDVI

What the app does. Three models share the same controls. *Pooled:* yield = zone yield level + f(rate) + controls, one curve for the field with an intercept per original zone (the flat-rate optimum). *Zone:* yield = f_zone(rate) + controls, rate terms per analysis zone and one intercept per original zone nested inside it. *NDVI:* yield = f(rate) + g(rate) × NDVI + controls, the quadratic coefficients linear in the continuous multi-year NDVI. Yield levels always come from the original zones, so merging zones for the curves does not merge their levels (otherwise a level gap was fitted as curvature; on a synthetic trial with one true response the merged optimum moved from 130 to 90 or 152 with a 15 bu/ac gap). The principle: VR pays when the response differs between parts of the field, not when yield level does.

What the literature says. The principle is well documented. Scharf 2006 (abstract), eight strip experiments with a quadratic-plateau per 20-m cell: yield level explained 15 % of the variance in the economic optimum; VR by yield level would have added $2/ha over uniform, VR by the local optimum $38/ha. Bullock 2002 (abstract): "failing to tailor input applications to site-specific response functions may misallocate inputs as much as uniform application does". Trevisan 2021 (full), four Illinois seed × N fields by mixed GWR: yield levels and optimal rates "are not necessarily correlated"; the varying intercept explained 25–50 % of yield variability, the local slopes about 3–5 %; its VR gain of $58/ha is in-sample. Kindred 2015 (full), three UK chequerboard wheat experiments: optima ranging by more than 100 kg N/ha within two fields, driven by soil N supply. Anselin 2004 (abstract, erratum) is the published analogue of the zone model: N × landscape-zone interactions with spatial errors, zone optima against a uniform one, weighted by field area per zone. Liu 2006 (abstract), optional, is the closest analogue of coefficients depending on continuous site covariates. Formerly cited here but not supporting the rule: Hegedus 2023 (full) fits no zone model and uses prior NDVI as an intercept covariate; Mieno 2024 (full, authors' manuscript source) is about functional-form bias in GWR valuation; Bullock 2000 (unread) does not state the principle in its abstract.

Support. Well supported for the principle; the triad itself, and multi-year satellite NDVI as the interacting covariate, are house design with no published counterpart among the papers read.

What changed after the check. "the value of VR depends on the response differing between sites, not on yield level" now cites Bullock 2002 and Scharf 2006, not Bullock 2000; the zone × rate interaction cites Anselin 2004. Mieno 2024 and Hegedus 2023 move to R1/E3; Bullock 2000 is kept only for the economics-of-information frame. Scharf 2006, Bullock 2002, Anselin 2004 and Liu 2006 are added; the three structures are stated as house design.

R3. Estimation — area-weighted least squares, controls, standard errors clustered by plot

What the app does. Each model is fitted by weighted least squares with the unit area as weight, with the other tested input (linear and quadratic, held at its status quo for prediction so a confounded second factor does not leak into the first), the combine shares (machine as a fixed effect) and a quadratic trend surface in

the coordinates as controls. Standard errors are clustered by plot, the unit of randomisation. Spatial correlation between neighbouring units is handled by the trend surface plus clustering, not by a geostatistical error model.

What the literature says. The premise, that independent-error inference on yield-monitor data is unreliable, is well supported. Marchant 2019 (full) requires a yield trend in the fixed effects and shows that mis-specifying its anisotropic covariance as isotropic shrank treatment standard errors by up to a factor of four. Griffin 2020 (full) finds the aspatial OLS misspecified, a soil covariate absorbing spatial structure. Alesso 2019 (abstract), 3,000 simulated fields: Type I error above nominal, rising with spatial autocorrelation, when it is not modelled. Paccioretti 2021 (abstract) compares three models that all carry a spatial error structure; no ordinary regression was fitted, so the earlier sentence "spatial-error models outperform ordinary regression" is withdrawn. The clustering level is supported by Abadie 2023 (full, arXiv text): "with clustered assignment, one should cluster the standard errors at the level of assignment"; this justifies the level, not the absence of a between-plot spatial model. The trend-surface route has a precedent in Lambert 2004 (abstract): a polynomial trend regression by OLS reached the same conclusion on N-response heterogeneity as the spatial econometric and geostatistical models (polynomial order not checkable). Cho 2021 (full) is the closest analogue of least-squares estimates with correlation handled only in the standard errors, but with a fitted spatial covariance, not plot clusters. Nothing read examines area weights, combine-share covariates or trend-surface-plus-clustering; Alesso 2021 (full) has no inference content and belongs to R4; Piepho 2011 (unread) cannot be scored beyond its abstract. The code docstring credits Bullock 2019 for "accounting for spatial structure"; that attribution was not checked in this round.

Support. Partly supported: well for the premise and for clustering at the randomisation unit; house choices for the area weights, the combine shares, the quadratic trend surface as the only spatial control, and plot clustering in place of a spatial error model (it ignores correlation across plot edges, so standard errors can still be optimistic on narrow plots).

What changed after the check. Abadie 2023, Lambert 2004 and Alesso 2019 are added; Alesso 2021 moves to R4; the Paccioretti 2021 sentence is reworded; Piepho 2011 is flagged as abstract only. The tests computed on these fits now use the F(h, G − 1) reference (Section 11, item 1) (see R5). The between-plot caveat is stated in this review; the app's PDF report does not yet print it.

R4. Zones without support are merged before the per-zone fit

What the app does. `merge_small_zones` merges a zone into its neighbouring NDVI class (the closer in mean NDVI when there are two) until every zone has at least 20 analysis units and the design rates its form needs: 3 distinct design rates for a quadratic or a plateau, 2 for a line. Because every original zone keeps its own intercept, only the rate variation inside each original zone identifies the curve, so a merged zone must also hold at least 2 design degrees of freedom (the sum over its original zones of distinct design rates − 1; 1 for a line): counting the union alone let {90} + {150, 170} pass with 3 rates while one degree of freedom set slope and curvature. `check_support` applies the same rule to every bootstrap resample and cross-fit half and raises instead of fitting, so a rank-deficient draw is skipped and counted rather than kept with an invented optimum.

What the literature says. No paper read gives a minimum number of units per zone, a minimum of 3 rates, or a merge rule. The principle that each sub-area must hold the full rate set has precedents: Scharf 2005 (abstract) fitted a quadratic-plateau to every 20-m section of field-length strips with discrete N rates from 0 to 280 kg N/ha; Kyveryga 2009 (abstract) replicated five N rates in 6–12 strips per field and found that classes by ECa, soil map unit and elevation explained under 3 % of the response variability. Alesso 2021

(full) is a design study (four simulated rates, 18 layouts, a linear response) whose only sufficiency statement is a GWR neighbourhood of about 3 % of observations chosen so every treatment was represented. Marchant 2019 (full) recommends at least 10 rows of the standard treatment and about 100 observations for a variogram; none of this is a per-zone rule. Li 2023 (working-paper text; journal version closed) found the quadratic coefficient of local regressions "highly sensitive to sample errors" even with six rates on 1,440 subplots and held it constant, so 20 units is a lenient floor for a free quadratic. Hernandez 2008 (abstract): 90 % intervals of the economic optimum spanned 29–106 % of its value, which supports reporting a thin zone's optimum with its interval rather than as a point.

Support. House convention. The 3-rate (2 for a line) condition is an identifiability requirement of the form, not a finding; 20 units is a house floor and lenient by the evidence of Li 2023.

What changed after the check. The thresholds are stated as house conventions. The old table's "300 OUs per zone (DIFM convention)" was never the app's value (default 20, range 5–300) and is replaced. The rate condition is documented as form-dependent, with the within-original-zone rule. Alesso 2021 moves here from R3; Scharf 2005, Kyveryga 2009 and Hernandez 2008 are added for the principle, not the numbers.

R5. The zone × rate test

What the app does. On the zone model, `_tests` runs one joint cluster-robust Wald test of H0 "the rate terms are equal in every zone". On the NDVI model it tests that the rate × NDVI terms are zero; on the pooled model that the rate terms are zero (no field response, used by E5). `_wald` judges the statistic against F(h, G − 1), h constraints, G plots (clusters); the chi-square p is stored as `p_chi2`. The verdict takes p < 0.10 as one of two conditions for VR, the other being the cross-fitted gain. No per-zone p-value is printed; no wild cluster bootstrap is implemented.

What the literature says. The machinery is Cameron 2015 (full), section VI: with few clusters "one should at a minimum use T(G − 1) critical values", joint tests should use F(h, G − 1), and even then "a test with nominal size 0.05 may have true test size of 0.10"; rejection rates were 0.063/0.058/0.080/0.115 at G = 50/20/10/6 in one study and 0.068/0.081/0.118/0.208 at G = 30/20/10/5 in another, worse with unbalanced clusters; the wild cluster bootstrap is "generally reliable" and preferred when G < 10. Anselin 2004 (abstract, erratum) is the precedent for a zone-dependent response: N × landscape-zone interactions with spatial errors, VR "modestly profitable" in all spatial models and in no non-spatial one; no formal equality test or alpha could be confirmed. Liu 2006 (abstract) backs the two-condition logic: clear site-specific response, yet site-specific N "would rarely cover its costs" on bootstrapped gain intervals. Trevisan 2021 (full) frames OFPE as the test of whether optimal rates vary enough to pay for VR, with local GWR t-tests, not a zone Wald test. Kyveryga 2018 (full) classes p > 0.10 as no evidence, 0.05–0.10 some, < 0.05 strong, and runs its examples at alpha 0.10. Formerly cited but without the test: Mieno 2024 (full, manuscript source) has no Wald test, clustering, alpha or decision rule, and says its plug-in bias also applies to "regression with interaction terms of field characteristics"; Bullock 2000 (unread) and Bullock 2020 (abstract) decide VR by economics or simulation.

Support. House convention: the pooled quadratic with zone × rate interactions, the joint Wald test, alpha 0.10, and requiring both the test and a positive cross-fitted gain. The small-sample reference is well supported.

What changed after the check. the three Wald tests are judged against F(h, G − 1) instead of chi-square (Cameron 2015); on the three delivered trials G = 82–168, so the change is small; it matters for small trials. The test and the cross-fit are no longer attributed to Mieno 2024, cited instead for the in-sample self-evaluation bias in E3; Bullock 2000 and Bullock 2020 leave R5; Anselin 2004 is cited for the interaction

model, Cameron 2015 for the inference, Kyveryga 2018 for alpha 0.10 as the on-farm convention (still a house choice), Liu 2006 and Trevisan 2021 for the two-condition logic. Multiple testing across zones (Benjamini–Hochberg, which Marchant 2019 used in its localised analysis) is noted as not applied: one joint test, no per-zone p-values. The wild cluster bootstrap for G < 10 is noted as not implemented.

7. Treatment strips

T1. Treatment comparison — RCBD on strip means, Tukey at alpha 0.05, on yield and net return

What the app does. When the design labels its plots by strategy (VR / Augmenta / Flat / None, or any labels), `engine/treatments.py` compares the treatments on yield and on partial net return with the plot (strip) as the experimental unit. With blocks it fits yield ~ treatment + block and runs Tukey–Kramer comparisons of the treatment effects with that model's residual mean square and degrees of freedom; without blocks, the one-way Tukey HSD. Alpha is 0.05. A one-way Tukey on a blocked trial kept the block variance in its error term and found no difference where the RCBD ANOVA found a strong one. Net return = crop price × yield − applied rate × input price (plus the VR cost for VR-labelled plots); a plot with no design rate and no as-applied value has an unknown cost and is left out of the net-return means and ANOVA, and the result says how many.

What the literature says. Design and unit are well supported. Kyveryga 2018 (full) defines the experimental unit as the plot or field-length strip treated independently, calls the RCBD "the most common experimental design for on-farm trials", and gives economic return as yield × price − input cost, computed descriptively. For more than two treatments it uses the least significant difference, with randomisation or paired t tests for two, at alpha 0.10, classing 0.05–0.10 as "some evidence". Marchant 2019 (full) fits an anisotropic mixed model to individual yield-monitor records, reaches treatment standard errors of 0.03–0.13 t/ha across layouts, and notes that farmers seldom recognise the 0.05 benchmark; Tukey, RCBD and multiple comparison do not occur in it. Lawes 2012 (abstract, corroborated by Cho 2021's description) found that a simple pairwise comparison of strips identified treatment differences as well as a spatial model, supporting a simple strip-level analysis. Alesso 2019 (abstract) supports replicated strip designs and the Type I warning for autocorrelated records analysed as independent; it does not say strip means are the remedy. Piepho 2018 (abstract) is the textbook-level reference for the mixed-model analysis of qualitative treatments, cited for the model, not the comparison procedure. Cho 2021 (full) is a single-strip method without replication. Piepho 2011 (unread) is cited by Cho 2021 for the strip-as-unit RCBD; nothing beyond its abstract is attributed to it. No source read uses Tukey; the procedure is Tukey 1949 (Crossref record only, no abstract).

Support. Partly supported: well for the strip as the unit, randomisation, replication and the RCBD; house choices for Tukey HSD at alpha 0.05 (the sources use LSD, randomisation or paired t at 0.10) and for the test on net return (descriptive in the sources).

What changed after the check. Tukey HSD now cites Tukey 1949 as the method; Kyveryga 2018 is described as using LSD and randomisation tests at alpha 0.10; Cho 2021 is kept only as the alternative for unreplicated single strips; Lawes 2012, Alesso 2019 and Piepho 2018 are added; Piepho 2011 is verified from its abstract only; the former "how many strips for a detectable difference" rule attributed to Marchant 2019 is withdrawn (the paper reports standard errors per layout, no rule). Alpha 0.05 and the net-return test are named as house choices; the Tukey–Kramer comparison on the RCBD model is documented in place of the one-way test.

Rule	Support	Verified references (keys)	House choices
R1 Response form	Partly supported; quadratic default is a house choice	Cerrato 1990 (abstract), Bullock 1994 (abstract), Miguez 2022 (abstract), Dhakal 2021 (abstract), Ackello-Ogutu 1985 (abstract)	Quadratic as default against the two classics; "plateau optimum less price-sensitive" as reasoning; line for two-rate designs; join point at or above the second-lowest design rate
R2 Three structures	Partly supported (principle well supported)	Scharf 2006 (abstract), Bullock 2002 (abstract), Anselin 2004 (abstract + erratum), Trevisan 2021 (full), Kindred 2015 (full), Liu 2006 (abstract, optional); Bullock 2000 (unread) frame only	The triad (field curve with zone intercepts, curve per zone, coefficients linear in NDVI); multi-year satellite NDVI as the interacting covariate; original-zone yield levels in the merged models
R3 Estimation	Partly supported	Marchant 2019 (full), Griffin 2020 (full), Alesso 2019 (abstract), Abadie 2023 (full), Lambert 2004 (abstract), Cho 2021 (full), Paccioretti 2021 (abstract); Piepho 2011 (unread)	Area weights; combine-share covariates; quadratic trend surface as the only spatial control; plot clustering in place of a spatial error model; other input held at its status quo
R4 Zone merging	House convention	Scharf 2005 (abstract), Kyveryga 2009 (abstract), Hernandez 2008 (abstract), Li 2023 (working paper), Alesso 2021 (full), Marchant 2019 (full)	20 units per zone (lenient); 3 design rates for a quadratic or plateau, 2 for a line; within-original-zone degrees of freedom; merge into the NDVI-nearest neighbour
R5 Zone × rate test	House convention; small-sample reference well supported	Cameron 2015 (full), Anselin 2004 (abstract + erratum), Liu 2006 (abstract), Trevisan 2021 (full), Kyveryga 2018 (full); Mieno 2024 (full) moved to E3	Joint cluster-robust Wald test of equal rate terms; F(h, G − 1) reference (Section 11, item 1); alpha 0.10; test plus cross-fitted gain both required; no per-zone p-values, no multiplicity adjustment, no wild cluster bootstrap
T1 Treatment strips	Partly supported	Kyveryga 2018 (full), Marchant 2019 (full), Lawes 2012 (abstract), Alesso 2019 (abstract), Piepho 2018 (abstract), Tukey 1949 (record only); Cho 2021 (full) alternative; Piepho 2011 (unread)	Tukey HSD / Tukey–Kramer at alpha 0.05 (sources: LSD, randomisation, paired t at 0.10); the test on net return; unknown-cost plots left out of the net-return comparison

8. Economics and the decision

This section covers the six rules that turn the fitted response curves into money and into a recommendation: the partial budget and the optimum (E1), the value of variable rate against its comparators (E2), the cross-fit that supplies the evidence (E3), the verdict on variable rate (E4), the choice of flat rate (E5) and the bootstrap intervals (E6). The rules are coded in `engine/economics.py` (`decide`, `crossfit`, `bootstrap`, `summarise_boot`) and `engine/project.py` (`verdict`); the wording below follows their docstrings. Every citation is a key of `references.csv`; after each key the text says how the paper was read for this check (full text, abstract, or abstract plus the author's own summary), because 13 of the 70 numbered references were judged from abstracts only and the maintainer asked that this be visible. Numbers attributed to a paper come from the verified evidence (`evidence_by_rule.json`), not from memory. Where a paper does not say what the rule needs, the text says so.

E1. Partial budget, the economic optimum and the field weighting

What the app does. Net return per acre is crop price × yield − input price × rate; costs that do not change with the rate are left out. The optimum of each curve (field, zone, or continuous NDVI) is the rate on a 161-point grid (121 points for the continuous NDVI prescription; 81 inside the bootstrap, the cross-fit and the price grid) at which this net return is highest, which for a smooth curve is where the slope of the response equals the ratio of input price to crop price. Optima are searched only inside the trial's rate range; an optimum at an end of the range is flagged as a corner, not reported as a measured optimum. Field-level values weight each zone by its acres inside the field boundary (the trial area when no boundary is given). A status quo outside the range is handled by E9, not here.

What the literature says. Bullock 2019 (full text) sets up the first half of the rule: the farmer maximises revenue minus input cost, max_x p f(x) − w x (its Eq. 1), and the optimum is where the slope of the response equals w/p (Eq. 2), derived separately for two soil zones; rate-invariant costs are absent by construction. It does not restrict optima to the trial range, does not weight zones by area and contains no grower-compensation computation. Mieno 2024 (full text) supplies the two clauses the earlier check had called house choices: an estimated optimum outside the trial rates is replaced by "the minimum or maximum trial N rate ... as the corner solution" (its example: 40 kg/ha becomes the lowest trial rate, 80 kg/ha), and the field profit of VR and of the uniform rate is the mean over 8,640 equal-area cells, which is area weighting; it attributes the standard four-step VR-versus-uniform evaluation to Swinton 1998 and Bullock 2002 among others. Swinton 1998 (abstract and extended summary) re-evaluates nine VR-fertiliser studies "using partial budgets adjusted to include minimum costs and grid cell areas", the partial-budget and area basis of this rule, and finds VR fertiliser unprofitable on wheat and barley, sometimes profitable on corn and profitable on sugarbeet. Bullock 2002 (abstract; body not read) carries the sentence the earlier text hung on Bullock 2000: unobserved soil characteristics "may" make an input's yield effect vary within a field, so failing to tailor inputs to site-specific response "may misallocate inputs as much as uniform application does". Mamo 2003 (abstract) is the strip-trial precedent: an EONR per sub-block, a uniform 145 kg N/ha that over-fertilised half the field, and VR by EONR that would have saved 69 and 75 kg N/ha and returned $8 and $23/ha more than the uniform rate in 1997 and 1999. Gardner 2021 (abstract) tests a zone prescription economically on a 31-ha Ohio field; Lambert 2006 (abstract) reports $28/ha over five years with sub-block aggregation; neither states the first-order condition or the costs omitted. Bullock 2000 and Lowenberg-DeBoer 1997 (abstracts only) support the production-economics frame and nothing more: neither readable abstract contains the formula, the marginal condition, a range restriction or area weighting.

Support. Well supported once re-cited. House choices: weighting by acres inside the field boundary rather than by trial-plot area; the 161-point grid (81 in the bootstrap, cross-fit and price grid); the 25 % rule for a far status quo (E9).

What changed after the check. The partial budget now cites Swinton 1998 and Bullock 2002 instead of the unread chapter Lowenberg-DeBoer 1997 and Bullock 2000, which stay in the general frame only and are marked as read from abstracts. Mieno 2024 was added for the corner-solution clipping and the area aggregation. Bullock 2019 moved from "does not support" to "partly" after the open-access full text was read; its compensation attribution was dropped (see E8). Every reference now states how it was read.

E2. The value of variable rate, its comparators and the ex post label

What the app does. Three strategies are priced on the same curves: the grower's status quo, the best single (flat) rate, and the zone rates. The value of VR is the net return of the zone optima minus the net return of the best flat rate, minus a per-acre VR cost the user enters (default 0); VR is also compared with the status quo. All of it is the value for this field in this season, i.e. ex post. The README's limits say that one season is ex post and that the grower should repeat the trial before acting on zone rates (E11). In the PDF report only the realised-trial table (and the trial's compensation cost) is labelled ex post, no ex ante recommendation is printed, and the advice to repeat the trial appears only with the weak ("may pay a little") VR verdict and as the caveat on a lower flat rate.

What the literature says. Trevisan 2021 (full text) is the published form of the three-way table: its Table 3 defines the status-quo net revenue r_sq = p f(N_sq, S_sq) − w_N N_sq − w_S S_sq, the optimal uniform plan as the argmax of the field-summed net revenue, the optimal site-specific plan per location and the gains of each over the status quo (the rates "the producers said they would have applied had they not taken part"); Table 5 reports per field the gain of the optimal uniform rate, of the optimal variable rate and the VR −

UR difference. It assumes "all equipment needed is already available at no extra cost", leaves information costs out, reports US$65/ha for optimised uniform rates and a further US$58/ha for site-specific rates, and asks how far one year of data can guide the next. Bullock 2020 (abstract; its 2013 precursor in full text) makes the cost deduction a separate decision step: the value of VR is defined "not considering the costs of the technology itself" (eqs. 3-8), and eq. 22 states "if the difference in the costs of the technologies is less than the difference in the net revenues ... adopt site-specific technology"; the precursor also separates the ex post question (what would have been best in the trial year) from the ex ante one. Its headline is a value-of-information result: on a "flat and black" Illinois field fifteen simulated years of OFPE did not make site-specific N profitable (H1 rejected), while a few years improved uniform-rate management (H2 not rejected). Swinton 1998 (abstract and extended summary) deducts study-specific minimum costs and concludes that profitability is site-specific. Bullock 2009 (abstract) frames VR value as the value of site-specific information: with that information free, VR nitrogen would have been profitable in six of eight fields, an ex ante estimate. Lambert 2006 (abstract) labels its $28/ha result "ex post"; whether a VRT cost was deducted is not stated. Rajsic 2008 (abstract) compares ex post optimal N with the constant ex ante recommendation on seven Ontario corn sites, the gap worth under $10/ha in about a third of trials; it computes no VR-versus-uniform value. Bullock 2000 and Lowenberg-DeBoer 1997 (abstracts) give the frame only; the sentence that VR value depends on how much the response differs between sites is not in Bullock 2000's abstract and could not be checked in its body.

Support. Well supported for the comparators, the separate cost deduction and the ex post label. House choice: a single user-entered $/ac VR cost (the literature either assumes equipment at no extra cost or re-budgets each study's minimum costs).

What changed after the check. Trevisan 2021 (Tables 3 and 5) and the Bullock 2020 eq. 22 decision rule are cited for the comparators and the cost rule; Swinton 1998 and Bullock 2002 replace Lowenberg-DeBoer 1997 and Bullock 2000 for the partial budget; the "value depends on response differences" sentence now rests on Bullock 2002's abstract, with its "may" kept; the ex post label cites Lambert 2006, Rajsic 2008 and Bullock 2020.

E3. Cross-fitting as the decision evidence

What the app does. In-sample, zone rates chosen on the data can never lose to one flat rate before the VR cost, so the in-sample gain and its bootstrap say how big the gain might be, not whether it exists. The evidence is therefore a repeated two-fold split of the plots, stratified by design rate (plots without a design rate form their own stratum): the zone optima and the best flat rate are chosen on the curves fitted to one half and priced on the curves fitted to the other half, then the halves swap roles. Each fold is judged only on the common applied-rate support of the two halves, [max(train low, test low), min(train high, test high)]; training optima are clipped into it, so the held-out curve is never extrapolated. The status quo is read on the held-out curve at its own rate, with the bounded linear extension of E9 outside the overlap. Thirty repeats are the default (the `crossfit` function's own signature default is 50). The summary works on the per-repeat average of the two folds: mean, standard deviation, 5th and 95th percentiles, the share of repeats with a positive gain and the number of repeats; there is deliberately no standard error, because the two folds of a repeat are complementary halves of the same plots and the 2R fold results are strongly dependent. Every skipped fold is counted by reason (no design support in a half, no rate overlap, a fit that failed), so the verdict can say why the value of VR was not tested instead of blaming the design. For the flat-rate decision of E5 the same machinery runs in a mixed mode: the flat rate is chosen on the training half by the single curve and valued, with the status quo, on the held-out zone model over the field's zone acres, the basis of the in-sample row it checks.

What the literature says. Mieno 2024 (full text) supplies the rationale and the sign. It names step (3) of the standard evaluation, assuming the estimated site-specific response functions are true, as "the primary source of bias", with functional-form misspecification only exacerbating it, and states that in-sample the site-specific rates "by construction ... out-perform" the uniform rate in every cell. In 1,000 Monte Carlo rounds the true value of a GWR-based VR strategy had a median of $0.81 to $7.83/ha with "nearly half of the true economic values ... negative", while the in-sample estimate had a median of $17.15 to $45.62/ha and was "positive in almost all rounds", a bias of up to about $40/ha; the conclusions generalise the bias to "whatever the underlying statistical methods". The paper proposes no cross-fitting, hold-out or repeated splits as the remedy (its only cross-validation selects a kernel bandwidth); it recommends flexible functional forms and simulation. The procedure comes from outside agronomy. Chernozhukov 2018 (arXiv text) defines cross-fitting as the sample-splitting procedure "where we swap the roles of main and auxiliary samples to obtain multiple estimates and then average the results", gives a K-fold version, defines the mean and median over S random splits with a variance that adds between-split variation, recommends medians and uses 100 splits in its applications. Athey 2016 (full text) is the "honest" estimation precedent: one sample builds the partition, another estimates the effects, because adaptive estimation puts "spurious extreme values" into the same leaf; non-honest 90 % intervals covered only 74 to 84 % in its abstract, and honesty costs 7 to 22 % in mean squared error. Hegedus 2023 (full text) judges net-return forecasts on held-out data, but the hold-out is a year (two years train, one tests) and one random 70/30 split, with RMSE as the metric; no VR-versus-flat gain, no fold swapping, no repeats. Li 2023 (proceedings full text; journal abstract) values estimated site-specific optima on the true simulated response, with the profit difference "always negative", over 1,000 rounds and thirteen designs; it is Monte Carlo with known truth, aimed at trial design, not a data split. Bullock 2020 (precursor full text) prices estimated rates on the true response, an oracle that real data do not offer. Gardner 2021 (abstract) is the agricultural precedent for testing a prescription on data that did not generate it, without sample splitting.

Support. Partly: the rationale is well supported, the procedure is a house construction built on Chernozhukov 2018 and Athey 2016. House choices with no published counterpart: two folds rather than K; stratification of the split by design rate; the common-support restriction; R = 30 (Chernozhukov 2018 used 100 splits); the mean rather than the recommended median; the 5-95 % range and the share of positive repeats as decision statistics.

What changed after the check. Chernozhukov 2018 and Athey 2016 are cited for the sample splitting; Mieno 2024 stays on E3 with the quoted passages and the bias numbers but is no longer credited with the cross-fit itself (the README line that attributed the zone × rate test and the cross-fitting to it was corrected); the sentence "the gain must be computed with cross-fitting" is now stated as the app's inference, not the paper's; Bullock 2020 is cited for value of information (E2, E11) rather than for the split; Hegedus 2023 and Li 2023 remain as precedents for valuing estimated optima on data they did not come from.

E4. The verdict on variable rate

What the app does. Variable rate is *supported* when three conditions hold together: the mean cross-fitted VR gain over the flat optimum of the zone model, after the VR cost, reaches the minimum worthwhile gain (default $3/ac); the gain is positive in at least 70 % of the repeats; and the zone × rate test (a cluster-robust Wald test of equal rate coefficients across zones, judged against $F(h, \text{plots} - 1)$) gives $p < 0.10$. It is *weak* when the mean gain reaches min(1, minimum/3) $/ac with at least 60 % of repeats positive and either the cross-fit condition of the supported tier (mean at least the minimum, at least 70 % positive) or the test ($p < 0.10$) holds; otherwise the verdict is *flat rate*, with the best flat rate from the single-curve model. A cross-fitted gain within $1/ac of the minimum is flagged borderline, but only when the in-sample and

cross-fitted gains agree in sign, because when they disagree the evidence is contradictory, not two equivalent choices. When the cross-fit was not run or every split was skipped, the verdict says VR was not tested rather than that the zones do not pay; with one analysis zone it says whether the field has one zone or whether its zones were merged for lack of data. The in-sample gain is printed beside the evidence and labelled "optimistic by construction".

What the literature says. Pannell 2006 (abstract, plus the author's own summary of the paper) is the rationale for a minimum worthwhile gain: in many decisions "even large deviations from optimal decisions make little difference to the payoff", with four consequences, among them that information used to refine decisions is often worth little and that precision-farming adjustments of input levels often deliver small benefits; the lime example in the author's summary has an optimum near 2.7 t/ha with every rate from 1.8 to 4.3 t/ha within 5 % of the maximum payoff. It gives no dollar threshold and no probability or p-value criterion. Pannell 2019 (abstract) ties flatness to VR directly: the flatter the payoff function, the less a farmer gains from varying rates across a field; IR95 and relative curvature are found wanting and the standard deviation of site-specific payoff slopes at the optimal uniform rate (SDS) is proposed as a measure "highly correlated with the benefits of variable-rate management". Lowenberg-DeBoer 2019 (full text, author's open copy) records that VRT adoption "now rarely exceeds 20% of farms", which "suggests that farmers like the idea of VRT, but are not convinced of its value", and lists what VRT needs: lower cost, decision rules giving "a high probability that the resulting application is the most profitable", and demonstrated value in the margin; the first profitability review (Swinton 1998) and most later studies questioned VR fertiliser economics. Mieno 2024 (full text) gives the reason small in-sample gains cannot be trusted (E3): true VR values with medians of $0.81 to $7.83/ha and nearly half negative against estimates of $17 to $46/ha. Lambert 2006 (abstract) reports +$28/ha over five years with no significance test, probability of a positive return or decision threshold. Lakens 2018 (full text of the author preprint) supports setting the smallest effect of interest before looking at the data, "ideally ... based on a cost-benefit analysis", and running an equivalence test and a significance test "in concert": an effect can differ significantly from zero and still be smaller than the smallest effect of interest. It is a psychology tutorial and the app does not run its TOST; it is cited for the principle. On the test component, the completeness check cites Cameron 2015 for the known over-rejection of cluster-robust Wald tests with few clusters (0.115 to 0.208 at nominal 0.05 with six clusters), which is why the test is judged against F(h, plots − 1) and why it is one condition of three rather than the gate. Lowenberg-DeBoer 1997 (abstract) supports only the premise that VR value must be shown field by field.

Support. Partly: each ingredient has published backing (a margin because payoff is flat; a high probability of profit; gains that are small, year-dependent and often negative; a smallest effect set a priori), the numbers do not. House calibrations, named as such: the $3/ac minimum worthwhile gain; the 70 % and 60 % shares; $p < 0.10$; the weak tier at min(1, minimum/3); the $1/ac borderline; the AND logic. They were set on one synthetic control (README, Validation): twelve simulated trials with a known answer, in which a chance rate imbalance between zones gave $p = 0.015$ and +$2/ac with no real zone difference, and after which the verdict was right in 6 of 6 trials with different zone responses (+$10.8 to +$16.5/ac cross-fitted) and 6 of 6 without (−$0.4 to +$0.7/ac). One control is a calibration, not a validation of the thresholds.

What changed after the check. Pannell 2019 and Lakens 2018 were added; Swinton 1998 replaces the 1997 chapter; the thresholds and the AND logic are now described as house calibrations on the synthetic control, with Pannell 2006 and Pannell 2019 cited for the rationale and not for the numbers; Mieno 2024 is cited for the bias, not for the procedure; the verdict's handling of a missing cross-fit and of a single zone was made explicit in the report.

E5. Which flat rate: the direction-aware rule

What the app does. The best flat rate replaces the status quo when its cross-fitted gain over the status quo is worthwhile (mean at least the minimum, positive in at least 70 % of the repeats) and its in-sample gain, computed on the main zone model over the field's zone acres (the basis of the summary table), is positive; a switch advised beside a negative printed gain would contradict itself. Direction decides the extra evidence. A *higher* rate has to buy its extra yield, so the field must also respond to the rate (pooled Wald test $p < 0.10$); a gain made of a chance slope is not advised. A *lower* rate saves input with certainty and loses only the yield the fitted curve gives up, so it needs no response test, but it must be vouched for by the cross-fit: with zero repeats the in-sample gain alone would advise the cut and the verdict refuses it. The reason text then decomposes the gain into the certain input saving, input price × (status quo − flat rate), and the yield term the curve credits or charges, and adds that under-application costs more than over-application when the true response is steeper than estimated (Babcock 1992; Rajsic 2008), so the grower should repeat the trial before cutting further. A gain within $1/ac of the minimum is flagged borderline only when every other condition of the switch holds. The rule was inverted before the eighth review round: the old rule kept the status quo and printed the opposite evidence beside it.

What the literature says. Pannell 2006 (abstract and author's summary) supports the gate: a status quo inside the flat range loses little, so a change must clear an economic margin; the author's summary adds that within that range considerations such as risk or liquidity can guide the choice of rate, the only published hint of a basis for preferring the cheaper rate inside a flat range. Nothing in the abstract or the summary discusses significance tests or a cut-versus-increase asymmetry; the full text was not readable, so the earlier statement that "the paper does not discuss" asymmetry was reworded to what was read. Pannell 2019 (abstract) adds that near any economically best management there are other options that perform almost as well. The asymmetry in the literature runs the other way from a naive "the cut is safe" reading. Rajsic 2008 (abstract) finds the fixed ex ante recommendation above the ex post optimum on most sites, but where it falls below, usually on less productive sites, the shortfall is much larger; because over-application costs little compared with under-application, the authors consider it reasonable to apply more than average to capture the good years; trials need a wide range of rates. Babcock 1992 (abstract) shows that for risk-neutral producers the optimal N rate may be higher under uncertainty about weather or soil N, as self-protection against being "caught short", contrary to the usual advice to cut N because it raises yield variance. The rule reconciles this by charging the fitted yield loss inside the gain of the cut, by requiring the cross-fit to vouch for it, and by printing the caveat; it does not treat the cut as costless. Altman 1995 (text via the abstract index) explains why a non-significant response test is the wrong gate for a cut: a P value above the conventional level does not mean nothing important was found; only 30 % of 71 trials with P > 10 % in Freiman's sample had a 90 % chance of detecting even a 50 % difference; associations should be quantified with an interval rather than judged by P alone. In this rule the H0 the pooled test cannot reject is exactly the case in which the cut pays, so the test cannot be its gate; the $p < 0.10$ screen for a higher rate is a conservative house screen, not evidence of no response. Lakens 2018 (full text) is cited for setting the $3/ac smallest effect of interest a priori, independent of the outcome. Bullock 2000 (abstract) contains nothing on replacing a status quo, on tests as gates or on asymmetry and was removed from this rule. No agronomic equivalence-margin paper was found (one Crossref query; an unread 2025 starter-P paper of unknown relevance).

Support. Partly, with the decisive parts house choices: the pooled $p < 0.10$ gate for a higher rate; the waiver of the response test for a lower rate and the "saving with certainty" accounting; the combination of a cross-fitted worthwhile gain and a positive in-sample gain. The literature supports the economic gate and the ex post caveat, and contradicts any reading of the cut as the safe error, which the report now says in so

many words.

What changed after the check. The direction-aware rule stays, with the cross-fit as the required voucher for a lower rate (mean at least the minimum, at least 70 % of repeats positive, in-sample gain positive). The report adds the Babcock 1992 and Rajsic 2008 caveat and the advice to repeat the trial before cutting further; Altman 1995 is cited for "absence of evidence is not evidence of absence" and Lakens 2018 for the a priori minimum; the reason text prints the input saving and the yield term of the cut. The evidence file further recommends printing the yield term with its bootstrap interval; the reason text currently prints its point value.

E6. Uncertainty of optima and gains: the plot bootstrap

What the app does. Plots are resampled with replacement (the plot is the randomisation unit and the cluster of correlated records), the models are refitted and the whole decision recomputed; the report gives, for each optimum and gain, the median, the 5-95 % percentile interval and the share of resamples with a positive gain. A resample that loses the design support of a curve (a zone of the full data missing; fewer than three distinct design rates in a zone or the field, two for a line; or too few design degrees of freedom inside the original zones) cannot identify its curvature and would return a pseudo-inverse or noise-fitted optimum, so it is dropped rather than fitted; the dropped and failed counts are printed. The default is now 500 resamples for a new project (the `bootstrap` function's own signature default is 200); 1,000 are advised for a report to publish; the three delivered trials were run with 100 and their intervals are labelled accordingly. The bootstrap describes the size of the gain, not whether it exists (E3).

What the literature says. Hernandez 2008 (abstract) is the agronomic precedent: four confidence-interval methods for the ex post optimum rate on a five-site dataset; 90 % intervals at four sites spanned 29 to 106 % of the optimum; "classical symmetrical CIs were not adequate", Wald intervals misplace both bounds, and the bootstrap methods "showed great potential"; the number of resamples is not in the abstract. Jaynes 2011 (abstract) computes EONR bands for seven response forms from "1,000 realizations", finds 68 % bands that "can span several tens of kilograms per hectare and are typically skewed", and concludes that "it is essential that confidence bands always be reported": wide and asymmetric, hence percentile rather than Wald. Field 2007 (abstract) gives the theory for resampling whole clusters: the cluster bootstrap is consistent under both the transformation and the random-effect model, where the residual bootstrap is not. Cameron 2008 (abstract) ties the bootstrap to the clustered inference of R3: cluster-robust tests "can over-reject ... with few (five to thirty) clusters", and cluster bootstrap-t procedures bring a 10 % rejection rate back to the nominal 5 %. Marchant 2019 (full text) supports reporting the uncertainty of profit-relevant estimates and respecting within-row correlation (standard errors below 0.05 t/ha when treatments vary along the row, the largest recorded 0.32 t/ha, isotropic models that "led to substantial under-estimation of the standard errors") and suggests using the fitted distribution for "the probability that the beta_j are positive or greater than a threshold that signifies profitability"; its uncertainty comes from REML mixed models, and the words bootstrap and resample do not occur. Two references cited before the check do not support the rule: Hegedus 2023 (full text) measures forecast error on held-out years, with zero occurrences of bootstrap or resampling, and Mieno 2024 (full text) never mentions a bootstrap or an interval on the gain, so the earlier sentence that it wants the gain "reported with bootstrap intervals" was unfounded. The textbook guidance on how many resamples a percentile interval needs could not be verified from a readable text and is not cited.

Support. Partly: the resampling unit (the plot as cluster) and the asymmetric percentile interval on the optimum are well supported; the number of resamples and the dropping rule are house choices. House: B = 500 by default (the agronomic precedent uses 1,000); dropping and counting resamples that lose design

support, for which no published counterpart was found.

What changed after the check. The default rose from 100 to 500 resamples, with 1,000 advised for a report to publish (Hernandez 2008; Jaynes 2011), and the three delivered trials carry a label saying their intervals come from 100; Hernandez 2008, Jaynes 2011, Field 2007 and Cameron 2008 replace Mieno 2024 and Hegedus 2023 on this rule, which move to E3; Marchant 2019 is kept for the within-row correlation and the probability-of-exceeding-a-threshold argument; the contradiction between the 500 of the parameter table and the 100 of the rule file is resolved by the new default; the dropped-resample rule is documented and its count printed; bootstrap medians are no longer printed as point estimates (eighth review round).

Summary of section 8

Rule	Support	Verified references (keys)	House choices
E1 Partial budget, optimum inside the range, field acres	Well (after re-citation)	Bullock 2019 (full text); Mieno 2024 (full text); Swinton 1998 (abstract, extended summary); Bullock 2002 (abstract); Mamo 2003 (abstract); Gardner 2021 (abstract); Lambert 2006 (abstract); frame only: Bullock 2000, Lowenberg-DeBoer 1997 (abstracts)	Weighting by field-boundary acres rather than trial-plot area; 161-point grid (81 in the resampling steps); 25 % rule for a far status quo (E9)
E2 Value of VR: status quo, best flat, zone rates, net of VR cost, ex post	Well	Trevisan 2021 (full text); Bullock 2020 (abstract; 2013 precursor full text); Swinton 1998; Bullock 2009 (abstract); Lambert 2006 (abstract); Rajsic 2008 (abstract); Bullock 2002 (abstract); frame only: Bullock 2000, Lowenberg-DeBoer 1997	A single user-entered $/ac VR cost
E3 Cross-fit as the evidence	Partly: rationale well, procedure house	Mieno 2024 (full text); Chernozhukov 2018 (arXiv text); Athey 2016 (full text); Hegedus 2023 (full text); Li 2023 (proceedings full text, journal abstract); Bullock 2020; Gardner 2021 (abstract)	2-fold split; stratification by design rate; common applied-rate support; R = 30; mean instead of median; 5-95 % range and share of positive repeats as statistics
E4 Verdict on VR	Partly: ingredients supported, numbers house	Pannell 2006 (abstract, author's summary); Pannell 2019 (abstract); Lowenberg-DeBoer 2019 (full text); Mieno 2024 (full text); Lambert 2006 (abstract); Swinton 1998; Lakens 2018 (full text); Cameron 2015 (test component); premise only: Lowenberg-DeBoer 1997	$3/ac minimum; 70 % / 60 % shares; p < 0.10; weak tier min(1, minimum/3); $1/ac borderline; AND logic; all calibrated on one synthetic control
E5 Direction-aware flat-rate rule	Partly / house: gate and caveat supported, asymmetric rule house	Pannell 2006 (abstract, author's summary); Pannell 2019 (abstract); Rajsic 2008 (abstract); Babcock 1992 (abstract); Altman 1995 (abstract index); Lakens 2018 (full text); removed: Bullock 2000	Pooled p < 0.10 gate for a higher rate; no response test for a lower rate, cross-fit as its voucher; crossfitted worthwhile gain AND positive in-sample gain; input-saving decomposition
E6 Plot bootstrap, percentile intervals	Partly: cluster unit and asymmetric interval supported, B and dropping rule house	Hernandez 2008 (abstract); Jaynes 2011 (abstract); Field 2007 (abstract); Cameron 2008 (abstract); Marchant 2019 (full text); do not support (moved to E3): Mieno 2024, Hegedus 2023	B = 500 default (1,000 advised; delivered trials at 100, labelled); dropping and counting resamples without design support

8. Economics and the decision (continued)

E7. Price sensitivity: the crop price × input price grid

What the app does. After the verdict, `economics.price_grid` re-runs the decision (`analyse`, 81-point rate grid) at every cell of a 7 × 7 grid of crop price × input price. Both axes span −30 % to +30 % of the prices entered in step 3 (`Prices.input_price_range_pct`, applied to both prices in `project.py`), so the

steps are 10 %. Each cell records the best flat rate, the zone optima, the VR gain over the best flat rate and the gain over the status quo; the PDF prints them as a pivot. The grid runs on the response form actually fitted (quadratic by default, quadratic-plateau as the sensitivity, R1); nothing is assumed about how that form reacts to prices.

What the literature says. Pannell 2006 (abstract, plus the author's own full summary of the paper; the publisher's text returned 403) is the reason a price-driven shift of the optimum matters little: payoff functions are flat near the optimum, so even large departures from the optimal decision change the payoff very little; in his lime example the optimum is 2.7 t/ha and any rate from 1.8 to 4.3 t/ha pays within 5 % of the maximum. Nothing in the abstract or the summary mentions prices, price ratios or a grid. Pannell 2019 (abstract only) links flatness to VR: a flatter payoff reduces the benefit of adjusting rates within a field, and the standard deviation of site-specific payoff slopes at the uniform optimum tracks that benefit; that prices scale those slopes is our inference, not the paper's statement. Kitchen 2010 (abstract only) is the one empirical case of a VR gain that moves with prices: in 16 corn N experiments (0–235 kg N/ha) the optimum varied by more than 100 kg N/ha within 13 of 16 fields, sensing earned about $25–50/ha, and its value rose as fertiliser cost rose relative to grain price; the abstract gives neither price levels nor a range. Bachmaier 2012 (full text) gives the mathematics by form: in the quadratic the optimum is where the slope equals r/p, so it and its confidence set move with prices; in the linear-plus-plateau the optimum sits at the transition and does not depend on prices unless r/p exceeds the rising slope. The paper uses one fixed ratio (r/p = 0.0054545: 1.20 EUR/kg N, 220 EUR/Mg grain) and runs no price sensitivity. Bullock 1994 (abstract only) bears on E7 only indirectly: the quadratic predicts an optimum 5–60 % above the quadratic-plus-plateau and reduces profit by $0.61–17.12/ha/yr, so the grid must be run on the form fitted. Rajsic 2008 (abstract only) says nothing about prices and does not support E7.

Support. Partly supported. The reason for the grid (flat payoff; a gain that depends on the price ratio) is published; the ±30 % span, the 10 % steps and the two-dimensional crop × input grid in place of a single price ratio are house choices (the 2-D grid gives the same optimum as the ratio but a different $/ac gain).

What changed after the check. The sentence "a plateau optimum sits at the join point, so it is price-insensitive" is true only for the linear-plateau (Bachmaier 2012, section 2.2); the quadratic-plateau the app fits has its optimum below the join point, where the slope of the quadratic segment equals r/p, so it still moves with prices, bounded above by the join. The sentence was corrected. Rajsic 2008 was moved out of E7 (it belongs to E5 and E11); Kitchen 2010, Bachmaier 2012 and Pannell 2019 were added; the ±30 % grid is named here as a house choice.

E8. The realised trial and the ex-post cost of the trial to the grower

What the app does. `economics.realised` takes the analysed units (plot × zone) and computes, for each, the net return actually obtained at the applied rate and what the same unit would have returned at the status quo, at the best flat rate, at the recommended flat rate and at its zone's VR rate. The counterfactual keeps the unit's own residual, y_cf = y_obs + ŷ(r_cf) − ŷ(r_applied): the model only moves the unit along its zone curve. The VR return is net of the VR cost; a status quo outside the trial range gets the bounded extension of E9, and one too far outside is not priced (`sq_too_far`). Area-weighted sums give the field differences (trial − status quo, VR − status quo, VR − flat) and a table by zone. With two inputs `project.realised_joint` computes the counterfactual once with every input at its status quo at once: the fitted yield is additive in the controls, so y_cf = y_obs + Σ_i (y_sq,i − y_obs); inputs without a priced status quo are named and left out. The PDF quotes the result once: the trial rates cost X $/ac against the status quo, × plot acres (extrapolated from the analysed acres) = total, "the ex-post cost of the trial to the grower, the quantity on-farm experimentation programmes compensate (DIFM pays a fixed amount per acre; Puntel

2024), computed here from the realised units"; within 0.5 $/ac the trial cost nothing, and a trial that out-earned the status quo leaves nothing to compensate.

What the literature says. Bullock 2019 (full text, re-read passage by passage) supplies the inputs only: cooperating farmers provide the "typical" way the field would be managed without a DIFM experiment (baseline seeding and fertiliser rates), which is the grower-declared baseline; plot interiors are cut into subplots of about 20 m that are the observational unit; the transitional buffer is not used; and the economics explicitly set aside the choice of several inputs "for simplicity". The word "compensation" occurs once, for yield-monitor flow compensation. Bullock 2020 (abstract only) does not support the rule: its benchmark is the MRTN recommendation, not a grower status quo, and its object is the ex-ante value of the information a trial produces, not the realised return of the trial as applied. Puntel 2024 (full text) supports the two things the DIFM papers do not: partnering farmers were provided with financial compensation for the cost and risk of the trial N plan, and the farmer's business-as-usual management was compared with the tested technology on a whole-field basis in N, yield, partial profit and NUE; no formula, residual mechanics or payment terms are given. Gardner 2021 (abstract only) is the closest framework: run the randomised experiment, estimate the response for the Rx's zones, then test whether the Rx is economically optimal, on a 2018 nitrogen × seed-rate field; its benchmark and cost conventions sit in the paywalled body. A local DIFM platform report shows that DIFM's own number is an estimated optimal-uniform versus grower-chosen difference on the predicted profit surface, not a realised accounting.

Support. House convention. The per-unit counterfactual with the residual kept, the area-weighted ex-post accounting, the one-cost-with-every-input-at-status-quo rule and the 0.5 $/ac "cost nothing" band have no published counterpart. Bullock 2019 backs the baseline and the unit, Puntel 2024 the practice of compensating and of scoring against business-as-usual, Gardner 2021 the framework.

What changed after the check. The label "DIFM grower compensation" was a misnomer and was replaced in the PDF by "the ex-post cost of the trial to the grower, the quantity on-farm experimentation programmes compensate (DIFM pays a fixed amount per acre; Puntel 2024), computed here from the realised units". Bullock 2020 is no longer cited for it; Bullock 2019 is cited only for the grower-declared baseline. With two inputs the cost is quoted once, from the joint counterfactual, in place of one sentence per input.

E9. A status quo outside the tested range; optima never leave it

What the app does. Optima are searched on a rate grid inside the trial range [lo, hi] only; an optimum at either end is a corner and is flagged in the app and the PDF. A status quo (or a recommended flat rate) outside the range is priced by `economics.bounded_curve`: inside the range the curve is itself; outside, the yield is extended linearly from the nearest edge with the curve's one-sided slope at that edge, never its curvature, and floored at 0; a plateau curve is extended the same way but never with a negative slope, and below the range with the slope of its rising segment. The input is charged at its own rate. `sq_position` measures how far the rate lies outside as a share of the range; beyond 25 % (`SQ_MAX_OUT`) the status quo is not priced at all, its columns are dropped and the PDF says so. The docstring records why the edge value is not simply held: that gave a status quo under the trial the yield of the lowest rate for less input, so it could never lose to that rate.

What the literature says. "Optima never leave the range" has published precedents. Scharf 2005 (full text) constrained the EONR never to exceed the highest N rate tested, 280 kg N/ha, reports EONR spanning 0 to 280 kg N/ha in five of eight fields, and replaced a fitted curvature the data did not support (positive quadratic coefficient → linear; negative linear coefficient → unresponsive; a fit that did not converge → linear; the

three cases together were 40 of 611 cells). Mieno 2024 (manuscript source and full text) uses the same convention in its simulations: where the estimated EONR falls below or above the trial rates, the minimum or maximum trial rate is used as the corner solution, and the trial rates are set 20 kg/ha beyond the quantiles of the true plateau points so that the range covers the optimum; a modelling practice, not guidance. Bachmaier 2012 (full text) gives the rationale for not using the fitted curvature outside the tested rates: the optimum estimate depends strongly on the N-level design, the area on which the model is fitted; the extreme levels dominate the regression through leverage; and the fit should be trusted in a neighbourhood of the optimum. The same section asks for designs that include zero and very high N, so it does not argue against wide ranges. Rajsic 2008 (abstract only) supplies the premise, the need for a sufficient range of rates in N field trials; Miguez 2022 (abstract only) adds that precision and accuracy of the optimum demand more rates and replications. Cerrato 1990 and Bullock 1994 (abstracts only) are functional-form papers: in Cerrato 1990 the optima run from 128 to 379 kg N/ha across five models at one price ratio; neither says anything about rates outside the tested range, and neither supports E9.

Support. Partly supported for the bound on optima and for the premise. The one-sided linear extension, the floor at 0, the 25 %-of-range cut-off and charging the input at its own rate while extending the yield are house choices with no reference.

What changed after the check. Cerrato 1990 and Bullock 1994 are no longer cited for E9 (they belong to R1); Scharf 2005, Mieno 2024 and Bachmaier 2012 were added; the extension rule, the floor and the cut-off are named here as house choices.

E10. The continuous-NDVI prescription as a diagnostic

What the app does. `economics.ndvi_prescription` evaluates the rate × NDVI model at 20 quantile bins of the field's pixels, each at the area-weighted mean NDVI of its own pixels (not the bin midpoint, because the end bins run to a slough or a yard), and chooses each bin's rate inside the trial range. Two flags follow. A bin whose mean NDVI lies outside the NDVI range of the analysis units is marked out of range, and its share of the field area (`share_out_of_range`) is printed and hatched in the figure: its optimum rests on the interaction extrapolated beyond the trial. A bin whose optimum sits at an end of the range is a corner (`share_corner`): with the interaction the curvature changes sign along the NDVI, so a convex or still-rising bin curve has no interior optimum and takes whichever end is cheaper. When more than 25 % of the field area is at a corner, or corners occur at both ends, the PDF says the NDVI-model gain is not a usable prescription and is reported as a diagnostic only. The zones, not the pixel model, carry the verdict.

What the literature says. Mieno 2024 (full text) supports the two halves that have a counterpart: the corner-solution convention quoted under E9, and distrust of fine-resolution prescriptions built from locally varying curvature. The coefficient on N^2 is estimated inaccurately, so the range of estimated EONRs is far wider than the true one; nearly half of the true economic values of GWR-based VRA are negative; the bias reaches about $40/ha; and the authors expect the bias for regression with interaction terms of field characteristics too. Colaço 2018 (abstract and introduction) reports that about 25 % of sensor-based N studies showed economic losses and that recommendations from empirical response curves cannot be confidently generalised beyond the conditions of the experiment. Colaço 2019 (abstract only): sensor calibrations pooled over 19 years of wheat reached an R^2 of about 0.25 and their benefit over a conventional rate was insignificant; univariate NDVI models have low accuracy. Piikki 2022 (abstract and highlights): the field-mean N rate comes from on-farm experiments and the vegetation index only adds to or subtracts from it, with a leave-one-trial-out error of 14 kg N/ha; a VI-driven rate is a modulation around an experimentally determined mean. Franzen 2016 (full text) does not support the rule: it catalogues four in-season sensor algorithms whose floors and caps are internal, and its core idea that yield potential and N

response are independent argues against an r*(NDVI) prescription rather than describing when to demote one.*

Support. House convention. The triggers (share of the field at a corner above 25 % or corners at both ends; bins outside the units' NDVI range), the 20 bins and the consequence "diagnostic only" have no published counterpart. The distrust of locally varying curvature and the diagnostic-only role of a vegetation index are supported.

What changed after the check. Franzen 2016 was dropped from E10; Colaço 2019 and Piikki 2022 were added. The rule text said "most of its bins"; the criterion stated above is the code's.

E11. One season is ex post

What the app does. Every gain is labelled ex post, this field and this season (E2); the Limits say the optimum of one year is not next year's recommendation and that the trial should be repeated before zone rates are acted on. The cross-fit (E3) protects against overfitting within the season, not against the next season; the price grid (E7) shows how much the answer depends on prices. A lower rate advised without a yield response carries the caveat that under-application costs more than over-application when the true response is steeper than estimated (E5).

What the literature says. Rajsic 2008 (abstract only; supports) is the source of the distinction: on seven Ontario corn sites over several years with four functional forms there was no statistical or economic basis for selecting one model; ex post optima varied widely between seasons while the ex ante recommendation was constant; the recommendation usually exceeded the optimum, but the few large gaps were under-recommendations on less productive sites; the profit gap was under $10/ha on about a third of the trials; and over-application costs less than under-application, so applying above the average to capture good years is reasonable. Puntel 2016 (full text) shows how much the season optimum moves: EONR 123–268 kg N/ha in continuous corn and 42–241 in soybean–corn (CV 22 and 32 %), spring precipitation the only significant predictor (R^2 0.27–0.45), and the determination is made after harvest, when yields are known. Mamo 2003 (abstract only) is the zone version: about half a Minnesota field responded to N, zone EONRs would have used 69 (1997) and 75 (1999) kg N/ha less and returned $8 and $23/ha more than a uniform 145 kg N/ha, and about 60 % of the field responded the same way in two years, so temporal variation must be weighed with spatial variation. Bullock 2020 (abstract only; partly) is a simulation on one low-heterogeneity Illinois field: fifteen years of OFPE did not make site-specific N profitable, while a few years improved the uniform rate against MRTN. Pannell 2006 and Pannell 2019 (abstracts; partly) give the flat-payoff reason the loss from an ex ante rate is usually small (E7). Lacoste 2022 (full text; partly) is the weakest: "plan, act, observe, reflect and repeat" and local replication are a participatory learning cycle, and the words ex ante, ex post and season do not occur in the paper; it is kept only for the on-farm-experimentation framing.

Support. Well supported. House choices: the number of seasons before zone rates may be acted on is unquantified in any source (Bullock 2020: a few versus more than fifteen, for one simulated field; Mamo 2003: 60 % of a field consistent in two of three years), and "loss usually small" is quantified for Ontario and Minnesota corn N and Western Australian lime, not for Prairie cereal seed or N rates.

What changed after the check. Puntel 2016 and Mamo 2003 were added; Lacoste 2022 was demoted to framing; the rule now says that the size of the ex ante loss is a corn and lime figure and carries the asymmetry of Rajsic 2008 (under-application is the costlier error) that E5 already uses.

9. Prices, units and Prairie context

P1. Urea 46-0-0, nutrient basis and bushel weights

What the app does. `engine/prices.py` holds the conversions. Every rate inside the engine, the PDF and the prescription is lb/ac in the input's basis: lb of product (seed; urea as applied) or lb of nutrient for an input priced per lb N. A product's nutrient fraction converts between the two: urea 46-0-0 0.46, anhydrous ammonia 82-0-0 0.82, UAN 28-0-0 0.28, ESN 44-0-0 0.44, MAP 11-52-0 0.52 (P2O5); the price per rate unit is the product price divided by the fraction when the rates are nutrient ($/lb N = $/lb urea / 0.46). Grain prices are entered per bushel and converted from $/tonne with the crop's bushel weight (wheat and durum 60, barley 48, oats 34, canola 50, peas and lentils 60, flax 56, corn 56, soybeans 60, rye 56 lb; 2,204.62 lb per tonne). A design or map in kg/ha is converted on loading (× 0.892179); an as-applied map that logs lb of product under a price per lb N is converted with the fraction, and step 1 asks what the map logs. The join checks the applied rates against the design rates and warns when a basis is wrong (J3).

What the literature says. None of this is a research finding. 46-0-0 is the guaranteed analysis of urea under the Fertilizers Act and Regulations (pure urea is 46.65 % N, so × 0.46 undercounts N by about 1.4 %, the label guarantee); $/lb N = $/lb urea / 0.46 is algebra; the bushel weights are Canadian Grain Commission conversion factors with no DOI-bearing source. Khakbazan 2013 (full text) is the Prairie paper that documents the practice: the non-coated urea was commercial-grade fertiliser containing 46 % N, Table 2 lists "Urea (46-0-0)" at $0.584/kg of product (ESN, 44 % N, $0.704/kg), rates are in kg N/ha (25/50/75, 20/40/60 and 30/60/90 by site), crop prices in $/kg (wheat 0.220, canola 0.495, barley 0.190), and N rate was the main variable explaining net revenue in wheat, barley and canola. Fast 2024 (full text) prices CWRS at $436/Mg and urea at $1,100/Mg of product (Table 3) with rates in kg N/ha and NET = Y × P − C × NR, but never states 46 %, 0.46 or a per-bushel price, and its printed equation leaves the product-to-N cost conversion unstated, so it cannot be cited for the conversion. Malhi 2001 (abstract, introduction and section snippets; body paywalled) is an N-management review with nothing on units or prices and does not support P1. Cerrato 1990 (abstract only) is cross-cited for the point that the optimum uses a fertiliser-to-grain price ratio in consistent nutrient units. Both Prairie economic papers price grain per tonne or per kg, not per bushel.

Support. House convention anchored on standards. The 46-0-0 and product-pricing parts are documented Prairie practice (Khakbazan 2013); the bushel weights, the nutrient fractions of the other products and the kg/ha factor are standards cited as such.

What changed after the check. Khakbazan 2013 replaces Fast 2024 and Malhi 2001 as the citation for the 46-0-0 sentence; Fast 2024 is kept for the Prairie net-return frame only; Malhi 2001 is dropped from P1; the 46-0-0 and bushel-weight standards are named as standards without a DOI.

P2. Prairie agronomy context

What the app does. P2 is interpretive, not computational. The app enforces no plausibility window; the context shapes how the report's text and the analyst read a result: a shallow seed-rate response in a cereal once the stand is adequate, an N response that plateaus, and in canola a response to both density and N. The defaults are malting barley about 300 seeds/m², a spring wheat and barley N plateau at 80–120 kg N/ha depending on moisture, and canola 100–150 seeds/m² at soil-test N.

What the literature says. Barley seed rate: O'Donovan 2012 (abstract only) tested 100–500 seeds/m² on AC Metcalfe over 24 site-years and found 300 seeds/m² usually optimal, with a risk of lower yield and plumpness above it. McKenzie 2005 (abstract only): 150–350 viable seeds/m² gave small yield gains, and maximum yield needed fertiliser plus available soil N above 31 kg N per Mg of yield, a yield-scaled rule, not a fertiliser band. O'Donovan 2011 (abstract only) tested 200 and 400 seeds/m² and N at 0–120 kg/ha with

yield still rising at 120: no plateau, and no 300 seeds/m². Wheat seed rate: Lafond 1994 (abstract only) found yield rising with seeding rate in all crops and years on a thin Black soil (wheat 34–202, barley 27–161 kg/ha) and related to the reciprocal of seeding rate; Collier 2021 (abstract only) found 400 better than 200 seeds/m² in ultra-early systems; Beres 2016 (abstract and Core Ideas) is winter wheat, where a treated 200 seeds/m² stand yielded like 400 and 400 returned −$11/ha; McKenzie 2011 (abstract only) is irrigated, with seeding rate mattering less than date. Wheat N: Fast 2024 (full text): 60–240 kg N/ha, 120 kg N/ha optimal for CWRS wheat when coupled with enhanced-efficiency fertilisers; Read 1982 (abstract only): on fallow in south-western Saskatchewan rates above 15 kg N/ha gave little gain and on stubble the return fell to 3.1 kg grain per kg N between 45 and 60 kg N/ha, so the Brown soil zone plateaus under 60; Campbell 1977 (abstract only): one-site lysimeters at two moisture levels, diminishing returns irrigated but a logarithmic curve dryland, dryland plants lost 20 % of their N above 61.5 kg N/ha; Mangin 2022a (abstract only): NUE of 60 % at high N and 21–36 % of uptake after anthesis, dependent on late-season rain. Mangin 2022 (abstract only) has no seeding-rate factor and does not support P2. Canola: Harker 2012 (abstract only): 150 versus 75 seeds/m² gave +0.07–0.16 Mg/ha and +50 % N up to +0.25 Mg/ha across eight sites; Harker 2017a (abstract only): 100 to 150 seeds/m² did not raise yield at soil-test N, and novel inputs did nothing; Harker 2017 (abstract only): only small seed responds over 50–150 seeds/m²; Angadi 2003 (abstract only): yield flat from 20 to 80 plants/m² in a normal year and declining below 40 in a dry one; Yang 2014 (abstract only): the density effect is strongest below 60 plants/m²; Malhi 2007 (abstract only): N 0–150 kg N/ha economics by cultivar on three Saskatchewan soils. Grant 2012 could not be read by any route; of Malhi 2001 only the abstract, introduction and section snippets could be read (see P1), and none of them states an 80–120 kg N/ha plateau.

Support. Partly supported. The 80–120 kg N/ha band and the spring-wheat seeding window are syntheses, i.e. house choices: no paper states the band, barley is still rising at 120 kg N/ha (O'Donovan 2011) and the dry zone plateaus under 60 (Read 1982).

What changed after the check. Two factual errors were corrected: O'Donovan 2011 never tested 300 seeds/m² (the figure is O'Donovan 2012), and Mangin 2022 has no seeding-rate factor. Read 1982, Campbell 1977, Lafond 1994, Angadi 2003, Yang 2014, Malhi 2007, Harker 2017 and Harker 2017a were added; the band and the windows are presented as defaults with the soil-zone caveat; Malhi 2001 and Grant 2012 are cited only for what their titles support. The Prairie NDVI–yield timing of Mkhabela 2011 (abstract only; MODIS at census-region scale: best prediction window late June to late July in the sub-humid zone, early July to early August in the semi-arid and arid zones) is cited in the report's zone text (Z1), not here.

Summary of E7–E11 and P1–P2

Rule	Support	Verified references (keys)	House choices
E7 Price grid	partly	Pannell 2006; Pannell 2019; Kitchen 2010; Bachmaier 2012; Bullock 1994 (form dependence only). Not supporting: Rajsic 2008	±30 % span; 10 % steps (7 × 7); 2-D crop × input grid instead of a price ratio
E8 Realised trial, ex-post cost	house choice	Bullock 2019 (baseline, unit); Puntel 2024 (practice); Gardner 2021 (framework). Not supporting: Bullock 2020	per-unit counterfactual with residual kept; area-weighted ex-post accounting; one cost with every input at status quo; 0.5 $/ac "cost nothing" band
E9 Status quo outside the range	partly	Scharf 2005; Mieno 2024; Bachmaier 2012; Rajsic 2008 and Miguez 2022 (premise). Not supporting: Cerrato 1990; Bullock 1994	one-sided linear extension; floor at 0; 25 %-of-range cut-off; input charged at its own rate

Rule	Support	Verified references (keys)	House choices
E10 NDVI prescription as diagnostic	house choice	Mieno 2024; Colaço 2018; Colaço 2019; Piikki 2022. Not supporting: Franzen 2016	20 quantile bins; corner share > 25 % or both ends; out-of-range flag; "diagnostic only"
E11 One season is ex post	well	Rajsic 2008; Puntel 2016; Mamo 2003; Bullock 2020; Pannell 2006; Pannell 2019; Lacoste 2022 (framing only)	number of seasons before acting unquantified; "loss small" from corn N and lime, not Prairie cereals
P1 Units and prices	house choice (standards)	Khakbazan 2013; Fast 2024 (net-return frame); Cerrato 1990 (price ratio). Not supporting: Malhi 2001	46-0-0 and bushel weights as standards without a DOI; nutrient fractions of other products; kg/ha × 0.892179
P2 Prairie context	partly	O'Donovan 2012; McKenzie 2005; McKenzie 2011; O'Donovan 2011; Beres 2016; Collier 2021; Lafond 1994; Fast 2024; Read 1982; Campbell 1977; Mangin 2022a; Harker 2012; Harker 2017; Harker 2017a; Angadi 2003; Yang 2014; Malhi 2007. Not supporting: Mangin 2022. Unread: Malhi 2001; Grant 2012	80–120 kg N/ha band; spring-wheat seeding window; canola window; none enforced by the app

10. House contributions with no published counterpart

The check separated what the app borrows from what it adds. The additions below are the app's own; they are tested on synthetic trials with a known answer (`tests/`), on sweeps where the method's error rate is measured (`tests/test_levels.py`), and on the three worked trials, but they have not been validated outside Olds College.

Where	House contribution	How it is validated today
Z1	Per season, the median of the three clearest dates with the highest field-mean NDVI inside 15 June – 15 August; relative normalisation; temporal mean over seasons	Mkhabela 2011 (window, MODIS scale); Lenoir 2024 (the Blackmore composite on Sentinel-2)
Z3	3 × 3 smoothing, equal-area tertiles, patches under 0.5 ha merged	Analogues in Ping 2003 and Javadi 2022 (post-classification smoothing, island removal); no counterpart for the numbers
J2, J3	95 % of the footprint inside one plot; units of ≥ 10 records and ≥ 150 m²; unit rate = median of the as-applied points	dos Santos 2025 (footprint methods change the optimum); Bullock 2019 (buffers); thresholds unreferenced
J4	Design levels read back from the as-applied map: 1-D optimal partition (Fisher 1958; Wang 2011) scored by separation, shape tests, small-design rule, mixed-plot rule above the map's noise	Sweeps: no wrong number of levels on designs of ≥ 20 plots per level, ~1 clean map in 150 refused; 90–94 % read on 5-plot-per-level designs; gradients on 8–20 plots slip through in 2–4 % of cases, always flagged (README, `tests/test_levels.py`)
R4	Merge a zone with < 20 units or < 3 design rates into its NDVI-nearest neighbour	Lenient against Li 2023 (quadratic term unstable even with 1,440 subplots); the merge is reported
R5	Zone × rate Wald test (cluster-robust, $F(h, G - 1)$) at $\alpha = 0.10$ as one of two conditions for VR	Anselin 2004 (zone × N interactions); Cameron 2015 (reference distribution); the role in the verdict is the app's
E4, E5	Minimum worthwhile gain \$3/ac, ≥ 70 % of repeats positive (60 % and min(1, \$3/3) for "weak"), zone × rate $p < 0.10$, AND logic; the direction-aware flat-rate rule	One synthetic control with no true zone difference ($p = 0.015$, +\$2/ac by chance) set the gate; Pannell 2006/2019 give the rationale, Lakens 2018 the practice of a pre-set smallest effect; six VR and six flat synthetic trials classified 12 of 12
E6	Plot bootstrap with resamples that lose design support dropped and counted	Field 2007 and Cameron 2008 for resampling the cluster; the dropping rule is the app's
E7	± 30 % price grid	Flat-payoff rationale (Pannell 2006); the span is a convention

Where	House contribution	How it is validated today
E8	Realised trial: each unit keeps its residual and is moved along its zone curve; the joint counterfactual with every input at its status quo	The quantity on-farm experimentation programmes compensate (Puntel 2024); the computation is the app's
E9	Status quo outside the trial range: one-sided linear extension with the curve's slope at the edge, never the curvature, floored at 0; not priced beyond 25 % of the range	Optima never leave the range as in Scharf 2005 and Mieno 2024; the extension and the cut-off are the app's
E10	Continuous-NDVI prescription reported as a diagnostic only when more than 25 % of the field has its optimum at a corner of the trial range, or corner optima fall at both ends; bins outside the units' NDVI range are flagged and their share printed	Colaço 2019; Piikki 2022; the corner-share rule is the app's
J2, C1	Swath-width check against the spacing of the yield passes; applied-vs-design ratio check for a basis or unit mix-up	Internal consistency checks; no counterpart found

11. What changed in the app and its documents after the check

1. **Reference distribution of the Wald tests.** The zone × rate, NDVI × rate and pooled response tests are judged against F(h, G − 1), G the number of plots (clusters), instead of chi-square (Cameron 2015, sect. VI: with G = 6 clusters a nominal 0.05 test rejects 0.12–0.21 under the chi-square reference). On the three worked trials G − 1 is 81–167 and the p-values move little. No verdict changed. It matters for small trials. The chi-square p is still stored (`p_chi2`) and the reference is printed with the test.
2. **Bootstrap size.** The default of new projects went from 100 to 500 plot resamples; 1,000 is advised for a report to publish (Jaynes 2011 drew 1,000 Monte Carlo realisations of the optimum; Hernandez 2008 backs bootstrap intervals on the optimum but its abstract gives no number of resamples). The three worked trials in this document were run with 100 and their intervals are labelled so in the PDF.
3. **Wording of the realised trial.** "The order of the DIFM grower compensation" became "the ex-post cost of the trial to the grower, the quantity on-farm experimentation programmes compensate (DIFM pays a fixed amount per acre; Puntel 2024), computed here from the realised units". Bullock 2020 is no longer cited for it.
4. **A lower rate advised without a response test.** The rule stays (Section 8, E5): the cut must be vouched for by the cross-fit. The reason printed with it now states that under-application costs more than over-application when the true response is steeper than estimated (Babcock 1992; Rajsic 2008) and tells the grower to repeat the trial before cutting further.
5. **Attributions corrected** in the README, the PDF's Methods page and the code comments: cross-fitting cites Chernozhukov 2018 and Athey 2016, Mieno 2024 the in-sample bias; the footprint overlay cites dos Santos 2025; yield-monitor delays cite Chung 2002; the NDVI window cites Mkhabela 2011; the partial budget cites Swinton 1998 and Bullock 2002; clustering cites Cameron 2015 and Abadie 2023; Tukey HSD cites Tukey 1949; Lee 2012, Gauci 2024, Bullock 2000, Malhi 2001 and Mangin 2022 are no longer cited for rules they do not support, and for the partial budget Lowenberg-DeBoer 1997 (only partly supportive, read from its abstract) gave way to its journal version, Swinton 1998.
6. **The defaults table** of the earlier literature base (5-fold cross-fit, 500 resamples, 300 units per zone, 50 % footprint share, a 4-SD outlier filter, a 5 %-of-field zone minimum) was replaced by the code's real defaults: 2-fold × 30 repeats, 500 resamples (100 on the worked trials), 20 units and 3 design rates per zone, 95 % footprint share, no outlier filter in OCVR (the map arrives cleaned), 0.5 ha patches.
7. **Access honesty.** Every reference in Section 12 says how far it was read. The PDF's reference page lists the 31 references its text cites and points here for the rest.

12. Verified reference list

See `docs/LITERATURE.md` (generated from `references.csv` and `verification_log.json`) for the list in Markdown; it is reproduced below.

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Appendix A. How the check was done

A.1 The rules. The 28 rules were written from the code and the README of OCVR 1.1.0 on 29 October 2026, each with its parameters as coded (`ocvr_rules.json`), and the reference numbers the 27 October literature base cited for it.

A.2 Reading and judging. For each rule one reader fetched every cited paper (the DOI landing page; the open-access full text through Unpaywall, publisher pages or repositories when available; otherwise the abstract from Crossref, OpenAlex or Semantic Scholar), judged each rule–paper pair as *supports*, *partly*, *does not* or *unread*, quoted or paraphrased the passage that bears on the rule, and proposed up to three further references, each confirmed to exist (DOI resolving on Crossref) and read at least at abstract level. A second, independent reader then tried to refute every judgement with the paper in hand, corrected pages, numbers and provenance, and returned the corrected judgement; where the two disagreed the refuter's reading, which was done later and with more of the text, stands. A final critic ranked the weakest rules, listed topics with no reference at all and the most valuable additions. The readers were Claude (Anthropic) language-model agents run in a 58-agent workflow with web access; a person (G. Siebert Filho) decided what to adopt and what to change in the app. Where a paper could not be read beyond its abstract, or not at all, the judgement says so and the reference carries that label in Section 12.

A.3 Verification of the references. `verify_dois.py` resolved every DOI of `references.csv` on Crossref (OpenAlex as the second source) and compared the returned title (token overlap ≥ 0.6), first author and year with the citation; 147 of 147 verified (one, Malhi 2007, by hand: OpenAlex spells the first author "Mahli"). The log is `verification_log.json`; the evidence per rule is `evidence_by_rule.json`.

A.4 Limits of the check. About 30 of the 54 originally cited references were judged from abstracts or publisher previews only, and four anchor papers not at all (Section 1); Section 12 labels how far each reference was read. The response fitting itself (clustered WLS, trend surface, the rate × NDVI interaction, the plateau optimiser) was checked for consistency with the literature but not re-implemented independently in another package. The Prairie-specific statements (P2) are syntheses of several trials, not findings of one. The sweeps that validate the level reading (J4) and the synthetic trials that calibrated the verdict (E4) are the app's own tests; no outside data set was used.

A.5 Files. `docs/literature/`: this document (`.md`, `.pdf`), `references.csv`, `verification_log.json`, `evidence_by_rule.json`, `verify_dois.py`, `update_refs.py` (writes the PDF's reference page and `docs/LITERATURE.md`), `build_pdf.py`.